

1 Highlights

2 **A BRAVE Alloy Design Campaign** 3 **(Bayesian Risk-aware Alloy discoVery and Exploration)**

- 4 • Risk-aware BO uses learned feasibility to explore phase-stability boundaries
- 5 • 48 HEAs from Al-V-Cr-Mn-Fe-Co-Ni-Cu synthesized over three closed-
6 loop iterations
- 7 • V drives both yield strength ($r=0.84$) and sigma-phase failure ($r=0.54$)
- 8 • V-rich FCC alloys reach 1480 MPa UTS at 50% elongation
- 9 • Optima cluster on KKT-active boundary. Hard filtering would exclude them

A BRAVE Alloy Design Campaign (Bayesian Risk-aware Alloy discovery and Exploration)

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Abstract

In constrained alloy optimization, the compositions with the highest performance potential often reside at the boundary of phase stability—where the risk of experimental failure is also highest. This work demonstrates this principle through a risk-aware Bayesian optimization campaign on single-phase FCC high-entropy alloys in the Al–V–Cr–Mn–Fe–Co–Ni–Cu system. A learned feasibility classifier, integrated directly into the multi-objective acquisition function, probabilistically penalizes candidates likely to produce failed experiments while preserving access to high-performing boundary compositions. From approximately 27,000

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CALPHAD-screened candidates, 48 alloys were synthesized over three closed-loop iterations targeting five objectives (yield strength, UTS/YS ratio, strain at UTS, dynamic-to-quasi-static hardness ratio, and simulated depth of penetration), exploring 0.12% of the feasible space. Two compositional regimes emerged: a V-rich, Ni-rich high-strength regime (UTS up to ~1480 MPa at 50% elongation) and a Mn-containing high-ductility regime (UTS/YS up to 4.20 at >50% elongation). Among feasible alloys, vanadium simultaneously drives yield strength ($r = 0.84$) and sigma-phase formation ($r = 0.54$ with infeasibility); at V = 24 at.%, the three strongest alloys and three sigma failures share the same compositional point. Additionally, the strongest performing alloys cluster in a narrow region of compositional space (V $\geq$ 20 at.%, Ni $\geq$ 36 at.%), representing ~100 of 27,074 feasible candidates—a probability of $P \approx 6.5 \times 10^{-6}$ under random sampling. This dual role—consistent with the KKT prediction that constrained optima lie on active constraint boundaries—required feasibility-aware acquisition to access; hard filtering would have excluded this region entirely.

Keywords:

High Entropy Alloys, Multi-objective Bayesian Optimization, Risk-aware acquisition, Feasibility awareness, CALPHAD, Quasi-static and Dynamic mechanical properties, Penetration resistance, Closed-loop alloy discovery

1. Introduction

Structural alloys for extreme environments—from ballistic armor to gas turbine blades—underpin performance in defense, aerospace, and energy systems [1–3]. Although significant advances have been made in computational screening, high-throughput synthesis, and machine learning, the alloy development cycle remains slow and resource-intensive [4]: typically, alloy development programs proceed through iterative synthesis–processing–characterization–testing loops, and each cycle demands reproducibility (i.e., multiple evaluations) across heats, processing routes, and testing protocols. This problem is compounded by the fact that the composition–process–microstructure–performance (PSPP) space over which alloy campaigns operate is vast, and only a minute fraction has been explored directly [5]. In high-entropy alloy (HEA) systems, where five or more principal elements define a combinatorially large design space, exhaustive exploration is impractical even with advanced experimental methods [6, 7].

Within this broad landscape, FCC HEAs present a rich but not sufficiently mapped

71 composition–property space. Strength and ductility are governed by competing de-
72 formation mechanisms—dislocation glide, twinning, and local chemical ordering—
73 whose relative contributions shift with composition in ways that binary or ternary
74 subsystem data cannot predict. Interaction effects among four or more solutes
75 (i.e., beyond pairwise or ternary interactions) are absent from lower-order assess-
76 ments, leaving large regions of the compositional space without reliable prop-
77 erty estimates. The strength–ductility trade-off, often treated as intrinsic, may
78 instead reflect constrained design trajectories; multi-component spaces can con-
79 tain regions where twinning-induced plasticity (TWIP), transformation-induced
80 plasticity (TRIP), or both enable simultaneous improvements [8–10]. Thermody-
81 namic constraints compound the challenge: CALPHAD databases show signif-
82 icant prediction errors at FCC stability boundaries in higher-order systems, and
83 compositions that appear promising from property models frequently form brittle
84 intermetallics. Mapping these composition–structure–property relationships thus
85 requires an exploration strategy that generates both optimized compositions and
86 the physical insight needed to interpret them.

87 To date, several distinct frameworks have been applied to this problem. Combi-
88 natorial and high-throughput (HTP) approaches, for example, can generate candi-
89 date compositions at scale, but open-loop strategies tend to produce a significant
90 fraction of infeasible compositions—alloys with undesired phase constitution—
91 making such campaigns inherently resource-inefficient [11, 12]. More targeted
92 campaigns, guided by Integrated Computational Materials Engineering (ICME)
93 models or domain expertise, reduce infeasibility risk but require substantial com-
94 putational investment and often struggle to integrate multi-scale simulations at ac-
95 ceptable computational cost [13–15]. Machine learning (ML) approaches com-
96 plement these frameworks by enabling rapid screening, yet they frequently defer
97 experimental validation to later stages, particularly for bulk structural alloys where
98 synthesis and testing costs are significant [16]. Despite the progress made, the
99 dominant bottleneck across all these approaches is resource allocation: deciding
100 which chemistries to interrogate under a finite experimental budget [3].

101 In this context, Bayesian optimization (BO) has rapidly emerged as a principled
102 approach to optimal resource allocation in materials discovery, framing the prob-
103 lem as sequential decision-making under uncertainty [17, 18]. In a BO frame-
104 work, probabilistic surrogate models (typically Gaussian processes [19]) quantify
105 epistemic uncertainty across the design space, and acquisition functions balance
106 (a) exploration of poorly characterized regions with (b) exploitation of promis-
107 ing candidates [3, 20]. Multi-objective formulations extend this framework to

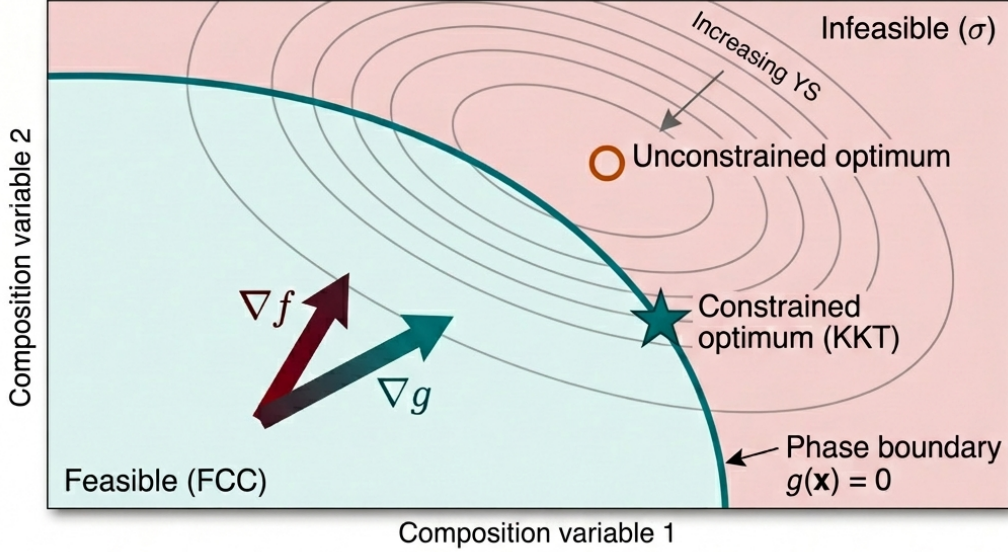
108 optimize competing performance metrics simultaneously, typically by maximiz-
109 ing the expected hypervolume improvement (EHVI) of the Pareto front [21–24].
110 Furthermore, metallurgical knowledge (thermodynamic stability limits, process-
111 ing windows, and microstructural descriptors) can be encoded directly into surro-
112 gate structure and constraint models, improving data efficiency while preserving
113 physical interpretability [25–27].

114 Over the past decade, closed-loop experimental campaigns that couple BO with
115 iterative alloy synthesis have validated this approach in practice. Xue et al. [28]
116 pioneered adaptive alloy design by identifying low-hysteresis NiTi-based shape
117 memory alloys through nine feedback loops exploring fewer than 40 composi-
118 tions from a space of 800,000 candidates. Rao et al. [29] applied iterative ML-
119 guided design to discover high-entropy Invar alloys with near-zero thermal expan-
120 sion. More recently, Sohail et al. [30] used domain-knowledge-informed active
121 learning to design FeNiCoAlTa alloys achieving 1.8 GPa yield strength with 25%
122 elongation. Additional closed-loop campaigns have targeted lead-free solders [31],
123 magnesium alloys [32], high-temperature shape memory alloys [33], and Al–Si al-
124 loys [34], with several incorporating multi-objective acquisition strategies [31, 35].
125 These studies demonstrate that BO-guided campaigns can identify competitive al-
126 loy compositions while exploring less than 1% of the feasible design space.

127 However, a practical constraint remains unaddressed in these campaigns: many
128 candidate designs are infeasible and yield no measurable objective values, wasting
129 experimental resources without contributing to surrogate model refinement. We
130 note that in alloy discovery, failure does not indicate poor performance (e.g., low
131 yield strength) but rather that experimental time and resources are expended with-
132 out obtaining usable measurements—because the alloy cannot be synthesized, pro-
133 cessed, or tested under the prescribed conditions. This distinction is consequential:
134 a low-performing alloy still provides training data for the surrogate model, whereas
135 a failed experiment provides none. The cost of each experimental iteration—
136 several thousands of dollars in materials, instrument time, and personnel—means
137 that even a small fraction of failed experiments can substantially reduce campaign
138 efficiency. Yet standard BO frameworks implicitly assume that every queried can-
139 didate returns measurable objectives, offering no mechanism to account for ex-
140 perimental viability during candidate selection. Figure 1 illustrates why this mat-
141 ters: in constrained optimization, the best solutions tend to reside on the feasi-
142 bility boundary—a pattern most cleanly expressed by the Karush–Kuhn–Tucker
143 optimality conditions [36] for continuous single-objective problems, and which
144 we invoke here as an intuitive framework for the discrete multi-objective setting

145 (formal statement in Section 3.4). A framework that excludes boundary compo-
146 sitions sacrifices the best candidates. One that ignores feasibility wastes budget
147 on infeasible ones. Although such boundary-adjacent optima have been observed
148 implicitly in metastability-engineered HEAs [9] and proposed algorithmically for
149 constrained BO [37, 38], they have not, to our knowledge, been demonstrated with
150 statistical significance in a closed-loop experimental alloy campaign—the gap the
151 present work closes.

(a) Constrained Optimum



(b) Three Strategies

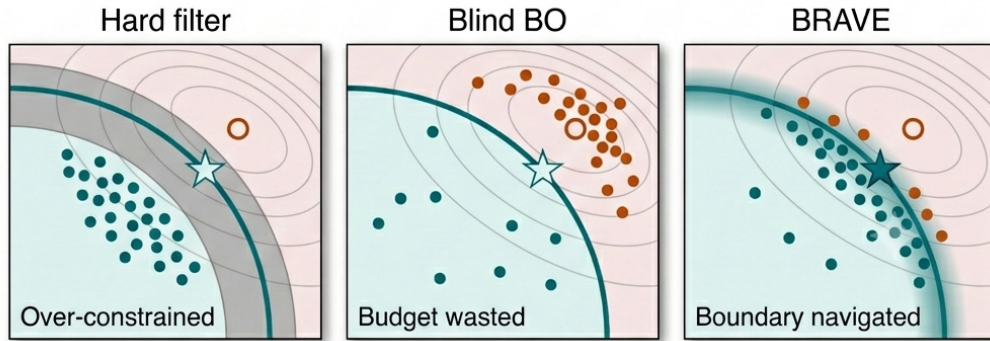


Figure 1. Constrained optimization in composition space. (a) When the objective (e.g., yield strength) increases toward the infeasible region (sigma phase), the KKT conditions place the constrained optimum (star) on the phase stability boundary where ∇f and ∇g are aligned; the unconstrained optimum (open circle) lies in the infeasible region. (b) Three acquisition strategies distribute experimental budget differently given the same total number of samples. Hard filtering over-constrains the search by excluding a buffer zone around the boundary, missing the KKT optimum entirely. Feasibility-blind BO concentrates samples near the unconstrained optimum, wasting budget on infeasible compositions. BRAVE navigates the boundary by concentrating samples in the feasible region near the phase stability limit, accepting a small number of infeasible outcomes as the cost of boundary exploration.

152 Constraint-aware acquisition strategies have begun to address this gap across sev-
153 eral domains. Hickman et al. [39] introduced feasibility-aware acquisition func-
154 tions that reduce failed experiments in perovskite and drug synthesis. Low et
155 al. [40] developed evolution-guided constrained optimization for self-driving chem-
156 istry laboratories. Chen et al. [41] coupled constrained expected improvement with
157 representation learning for computational catalyst and transparent conductor de-
158 sign. Tamura et al. [42] integrated feasibility constraints into autonomous robotic
159 materials exploration. In alloy systems specifically, constraint-aware formulations
160 have been proposed [25, 43], and physics-informed feasibility classifiers have been
161 developed [44], but no experimental alloy campaign has yet integrated a learned
162 feasibility model directly into the multi-objective acquisition function within an
163 iterative workflow.

164 The BIRDSHOT (Batch-wise Improvement in Reduced Design Space using a Holis-
165 tic Optimization Technique) framework was developed to integrate computational
166 screening, high-throughput experimentation, and batch Bayesian optimization within
167 an iterative design–make–test–learn workflow [45, 46] (Figure 2).

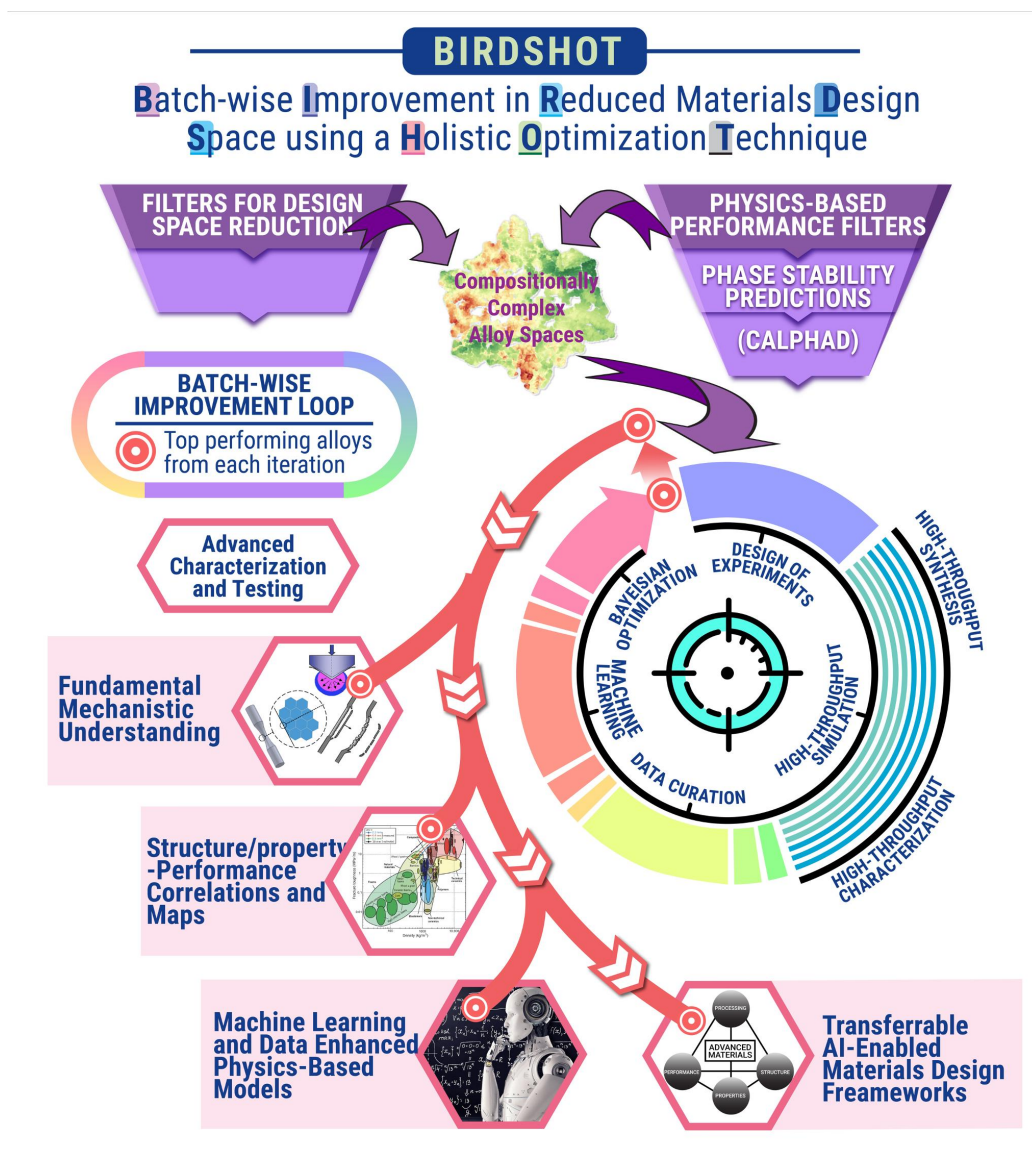


Figure 2. Overview of the BIRDSHOT framework. Physics-based filters (CALPHAD phase stability, property constraints) reduce the design space before optimization begins. The iterative loop cycles through design of experiments, high-throughput simulation, high-throughput characterization, data curation, and machine learning, with Bayesian optimization closing the loop at each iteration. Reproduced from: [46].

168 In a prior campaign, we applied BIRDSHOT to the six-component Al–V–Cr–Fe–

169 Co–Ni FCC HEA system, targeting three mechanical objectives (UTS/YS ratio,
170 hardness, and strain rate sensitivity) under four constraints (single-phase FCC at
171 700°C, density < 8.5 g/cm³, thermal conductivity > 5 W/(K·m), and SFE regime
172 classification) [46]. Five iterative design–make–test–learn loops, each synthesiz-
173 ing 8 alloys via vacuum arc melting, identified Pareto-optimal compositions by
174 exploring only 0.15% of the feasible design space. That campaign established
175 the viability of iterative, multi-objective Bayesian optimization for structural alloy
176 development. Although successful in identifying Pareto-optimal compositions, it
177 also exposed limitations. CALPHAD phase stability predictions were imperfect:
178 several alloys predicted to be single-phase FCC showed sigma or BCC secondary
179 phases. The framework had no mechanism to account for these failed experiments,
180 which consumed resources without contributing to model refinement. A detailed
181 summary of this earlier work is provided in the Supplemental Information.

182 In the present work, we report a second discovery campaign that extends the BIRD-
183 SHOT framework in three directions, addressing the limitations identified above.
184 The campaign also exposes a tension inherent to this alloy system: vanadium, the
185 dominant strengthening solute, simultaneously promotes sigma-phase formation,
186 forcing the best alloys onto the FCC phase stability boundary where the risk of
187 experimental failure is highest.

188 First, we introduce a risk-aware acquisition strategy—Bayesian Risk-Aware Alloy
189 Discovery and Exploration (BRAVE)—that penalizes candidates likely to produce
190 failed experiments (i.e., alloys that form secondary phases and yield no usable
191 objective data). Alloys predicted to form single-phase FCC structures but found
192 to contain secondary phases are classified as failures: they introduce processing
193 challenges and do not provide a consistent basis for comparison within the intended
194 single-phase design space. To mitigate this risk, we extend the EHVI acquisition
195 function with a learned Gaussian process classifier that estimates the probability
196 of experimental viability for each candidate composition. High-risk candidates
197 receive a probabilistic penalty during acquisition scoring, reducing their selec-
198 tion priority. We note that *the penalty is soft*: a candidate with sufficiently high
199 predicted objective values can still be selected despite elevated risk, preserving
200 controlled exploration in uncertain but potentially high-performing regions.

201 Second, we expand the design space from the previous 6-element system to an
202 8-element system, Al–V–Cr–Mn–Fe–Co–Ni–Cu. The inclusion of Mn and Cu
203 introduces additional degrees of freedom for tuning ductility and phase stability,
204 guided by prior experimental insights and domain knowledge. Mn is expected to

205 reduce the stacking fault energy in FCC alloys, potentially promoting mechan-
 206 ical twinning and enhancing strain-hardening capacity. Cu expands the accessible
 207 composition space but has positive mixing enthalpies with Fe, Co, and Cr, driv-
 208 ing FCC phase separation into Cu-rich and Cu-lean regions and creating thermo-
 209 dynamic boundaries that test the limits of current CALPHAD predictions. Grid
 210 sampling at 4 at.% resolution yields approximately 3.4 million candidate compo-
 211 sitions, of which $\sim 27,000$ (i.e., less than 1%) are predicted to form single-phase
 212 FCC structures after CALPHAD screening. To enable tractable exploration of this
 213 feasible space, we use a diversity-aware sampling strategy based on k-medoids
 214 clustering [27] across 37 chemically distinct subsystems. This prioritizes com-
 215 positionally informative candidates over redundant clusters, improving early-stage
 216 surrogate learning through better coverage of the design space.

217 Third, five objectives are evaluated through a coordinated high-throughput charac-
 218 terization workflow: yield strength (YS), UTS/YS ratio, strain at UTS, dynamic-
 219 to-quasi-static hardness ratio ($H_{\text{dyn}}/H_{\text{qs}}$), and depth of penetration (DoP). These
 220 objectives serve as complementary probes of deformation behavior across strain
 221 rates, spanning (a) quasi-static tensile response, (b) rate-dependent behavior mea-
 222 sured by nanoindentation and split Hopkinson pressure bar (SHPB) testing, and
 223 (c) simulated impact resistance computed via finite-element modeling. Among
 224 these, DoP is derived from ballistic impact simulations parameterized by Cowper-
 225 Symonds constitutive models calibrated to the quasi-static tensile and dynamic
 226 SHPB data acquired within each iteration. This creates a simulation-in-the-loop
 227 architecture in which experimental measurements from one stage of characteriza-
 228 tion feed directly into a computational objective evaluated within the same design
 229 cycle, coupling experiment and simulation within a single design iteration. Al-
 230 though DoP correlates strongly with YS ($r = -0.94$; Section 3.3), the calibrated
 231 Cowper-Symonds exponent encodes rate-sensitivity signal invisible to quasi-static
 232 testing alone—a distinction quantified in Section 3.3.6—so DoP contributes infor-
 233 mation beyond that carried by YS.

234 The complete BIRDSHOT framework—encompassing computational design, syn-
 235 thesis, characterization, and simulation—is summarized in Figure 3. By address-
 236 ing thermodynamic constraints, manufacturability, and experimental throughput
 237 within a closed-loop workflow, this campaign demonstrates risk-aware, multi-objective
 238 optimization through three iterative experimental cycles. The remainder of this
 239 paper presents the design methodology, experimental results across three closed-
 240 loop iterations, and the implications for feasibility-aware optimization in high-
 241 dimensional compositional spaces.



Figure 3. BIRDSHOT workflow from computational design through synthesis, characterization, and simulation. The iterative loop feeds experimental measurements and phase verification results back into the Bayesian optimization framework, which updates surrogate models and the feasibility classifier before selecting the next batch of candidate alloys.

2. Methods

2.1. Alloy Design Methodology

2.1.1. Design Space Definition and Compositional Constraints

The design space is an 8-element HEA system, Al–V–Cr–Mn–Fe–Co–Ni–Cu, extending the 6-element Al–V–Cr–Fe–Co–Ni system investigated in Campaign 1 [46] by the addition of Mn and Cu (rationale discussed in Section 1). The system includes all subsystems from 5 to 8 elements. Unlike Campaign 1, which imposed four constraints (single-phase FCC, density, thermal conductivity, and SFE regime), this study enforces only phase stability (i.e., single-phase FCC across the designated temperature range), adopting an open-ended strategy that avoids prematurely excluding regions of the composition space.

The initial compositional space is discretized at a resolution of 4 atomic percent, resulting in a pool of 3,365,848 unique compositions. To ensure manufacturability and focus on regions with potential for high performance, expert recommendations were used to impose upper limits on elemental fractions: $\text{Al} \leq 0.24$, $\text{V} \leq 0.24$, Cr

257 ≤ 0.50 , $\text{Mn} \leq 0.40$, $\text{Fe} \leq 0.50$, $\text{Co} \leq 0.50$, $\text{Ni} \leq 0.60$, $\text{Cu} \leq 0.24$. These com-
258 positional bounds reduce the design space to 1,504,938 candidate alloys, of which
259 1,490,129 were successfully evaluated using CALPHAD due to solver limitations
260 encountered on the high-performance computing cluster.

261 2.1.2. Candidate Screening via CALculation of PHase Diagram (CALPHAD) Pre- 262 dictions

263 To enforce the FCC phase stability constraint, we screened the reduced composi-
264 tional space using CALPHAD simulations. Thermo-Calc demonstrated effective
265 single-phase predictions for the Al-V-Cr-Fe-Co-Ni system in Campaign 1 [46].
266 Using the TCHEA6 database, we retained compositions with FCC phase fraction
267 ≥ 0.99 at 700°C, reducing ~ 3.36 million candidates to a feasible space of 26,621
268 alloys (approximately 0.8% of the original design space). Visualizations of this
269 downselection are included in the **SI**.

270 Phase prediction discrepancies in the first optimization iteration (BBB)—alloys
271 predicted as single-phase FCC that experimentally formed sigma—motivated a re-
272 analysis using the updated TCHEA7 database, released during this study. TCHEA7
273 enabled prediction of multiple phase regions (FCC, BCC, HCP, SIGMA, and liq-
274 uid) across different temperature regimes. We reevaluated the design space at
275 both 700°C and 950°C, advancing only alloys that retained FCC under both condi-
276 tions. This dual-temperature screening yielded a final feasible set of approximately
277 27,240 candidate alloys.

278 2.1.3. Sampling and Candidate Selection

279 Given the large number of candidate alloys resulting from the initial filtering and
280 screening (over 26,000), an efficient sampling strategy was critical to balance ex-
281 perimental feasibility with adequate coverage of the design space. For iteration
282 BBA (zeroth iteration), sampling was performed over the initial TCHEA6-screened
283 feasible space of 26,621 alloys (the TCHEA7 re-screening to 27,240 alloys oc-
284 curred after iteration BBB). We implemented a hierarchical sampling strategy us-
285 ing k-medoids clustering. First, the entire feasible space was segmented based on
286 chemical composition into 37 distinct subsystems, defined by the specific combina-
287 tion of elements present (for example, alloys containing all eight elements formed
288 one subsystem, while seven-element alloys were grouped by which element is ab-
289 sent). Details of subsystem grouping are provided in **SI**.

290 Subsequently, we applied k-medoids clustering to the feasible space to extract a
291 subset of 1000 representative alloys. In this clustering process, we enforced a con-

292 straint that at least one candidate alloy was selected from each of the 37 subsystems,
293 with the number of selections per subsystem proportional to the number of feasible
294 candidates available within that system. This approach guaranteed broad chemical
295 coverage and preserved the diversity of the design space.

296 From this subset of 1000 alloys, a further down-selection identified 16 candidate
297 alloys for synthesis. The downselection maximized representation across all 37
298 subsystems, ensuring that the final batch covered as many distinct chemistries as
299 possible and provided a well-distributed initial training dataset for the Bayesian
300 optimization loop.

301 2.2. *Synthesis*

302 The computationally designed alloys were synthesized using high-purity elemental
303 (> 99.9 wt.%) precursors to produce ~ 35 g dense, crack-free ingots using a
304 Vacuum Arc Melting (VAM) (Edmund Bühler AM 200) system. The elements
305 were initially melted in Cu crucibles under a high-purity Ar atmosphere. Each
306 ingot was flipped and re-melted 10 times to promote overall uniform chemical ho-
307 mogeneity. Mass loss was monitored throughout processing, and ingots showing
308 losses greater than 0.5% were reprocessed. Post-solidification, the ingots were ho-
309 mogenized between 950°C to 1150°C temperature ranges for 24 h in a Centorr LF
310 Series Model 22 high-temperature furnace. The homogenization heat treatment
311 temperatures were selected based on the predicted solidus temperature. Prior to
312 heating, the furnace chamber was purged with Ar gas multiple times to achieve a
313 vacuum of $\sim 5 \times 10^{-5}$ torr. The heat treatment was conducted under a low-pressure
314 Ar atmosphere ($< 10^{-2}$ torr), followed by furnace cooling.

315 The homogenized ingots were subsequently cold rolled to a $\sim 60\%$ reduction in
316 thickness. For recrystallization, the rolled specimens were wrapped in stainless-
317 steel foil with Ti sponge sacrificial getter to minimize oxidation. Recrystalliza-
318 tion was performed at 950°C for 30 min, followed by water quenching to suppress
319 secondary-phase precipitation during cooling.

320 Test sample profiles were precision-cut using wire-Electrical Discharge Machining
321 (EDM). Approximately 4 miniaturized flat tensile specimens, modeled following
322 modified SS-3 designs, were prepared for quasistatic tension testing. Each tensile
323 specimen had a total length of ~ 26 mm, a gauge length of ~ 8 mm, and a typical
324 cross-sectional area of 3×1 mm², with the loading axis aligned parallel to the
325 rolling direction. Prior to testing, the tensile samples were ground using 120 grit
326 SiC abrasive papers to ensure grip section parallelism, and pinholes were drilled

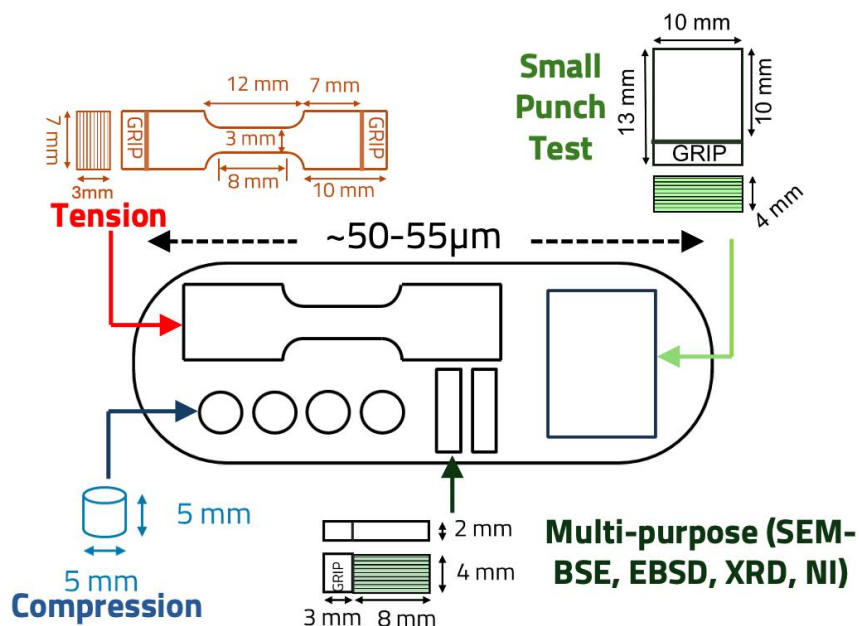


Figure 4. Representative schematics of ingots post-cold rolling with corresponding approximate sample dimensions for distinct characterization tasks: SEM/EBSD, EDS, XRD, and mechanical (Vickers, Nanoindentation) analyses, quasistatic tensile deformation, and Kolsky-bar compression dynamic mechanical response tests.

327 using carbide drill bits. Additional specimens were prepared for microstructural
 328 and mechanical characterization, as follows: two specimens, each ~2 mm thick
 329 with an exposed face area of 8.4 mm² normal to the long rolling direction, were
 330 designated for nanoindentation. Another two samples, 1.2 mm thick with a face
 331 area of 14.4 mm² (also normal to the rolling direction), were allocated for scan-
 332 ning electron microscopy (SEM), electron backscatter diffraction (EBSD), energy-
 333 dispersive X-ray spectroscopy (EDS), X-ray diffraction (XRD), and Vickers hard-
 334 ness (HV) measurements. A schematic of the specimen geometries is shown in
 335 [Figure 4](#).

336 We note that manganese contents exceeding 20 atomic percent introduced signifi-
 337 cant challenges during arc melting. Mn has a relatively low boiling point and high
 338 vapor pressure compared to the other constituent elements, making it volatile under
 339 the localized temperatures generated during arc melting. This volatility leads to
 340 Mn evaporation, resulting in deviations from the nominal composition and causing

341 microstructural inhomogeneities and porosity in the cast ingots. Mn loss disrupts
342 the stability of the intended single-phase FCC structure and promotes the forma-
343 tion of intermetallic compounds or segregated phases.

344 2.3. Verification

345 2.3.1. Metallography

346 Microstructural characterization and nanoindentation samples were prepared by
347 mechanical polishing using a Buehler AutoMet 250 autopolisher with diamond
348 suspensions down to 1 μm , followed by vibratory polishing in 0.04 μm colloidal
349 silica for EBSD analysis. A high-throughput sample preparation workflow was
350 developed over the course of the campaign, evolving from individual mounting (8
351 samples per batch) to large-format aluminum platens supporting 16 or more sam-
352 ples simultaneously, with laser marking for traceability. Details of this workflow
353 evolution are provided in the SI.

354 2.3.2. Chemical, Microstructural, and Phase Analyses

355 Compositional analysis was performed using a Tescan FERA-3 Model GMH Fo-
356 cused Ion Beam Microscope equipped with an Oxford energy-dispersive X-ray
357 spectroscopy (EDS) detector, operated at an accelerating voltage of 20 kV. EDS
358 point measurements were obtained from three spatially distinct regions of each
359 processed ingot, with five acquisition points per region and an integrated time of
360 ~ 50 s per point. The overall composition of each ingot was determined as the
361 arithmetic mean of the 15 measurements. Inter-regional variability was evaluated
362 to assess chemical homogeneity. Representative EDS results from three indepen-
363 dent synthesis iterations are presented in [Figure 5](#). The mean deviation between
364 measured and nominal compositions was ~ 0.42 at.%, with 99% of individual el-
365 emental measurements deviating by less than 1 at.%, confirming compositional
366 uniformity across all synthesis batches.

367 Microstructural characterization of the undeformed samples was performed by
368 backscattered electron (BSE) imaging at 15 kV using a Phenom XL scanning elec-
369 tron microscope (SEM) with a working distance of ~ 6 mm. Electron backscatter
370 diffraction (EBSD) measurements were carried out using a FERA SEM equipped
371 with an Oxford Symmetry CMOS-based EBSD detector, operated at an accelerat-
372 ing voltage of 20 kV, a beam current of 18 nA, and a specimen tilt angle of 70° .
373 Crystallographic orientation (IPF) maps were acquired on surfaces normal to the
374 cold-rolling direction over an area of approximately $1 \times 1 \text{ mm}^2$, using a step size
375 of $\sim 2 \mu\text{m}$.

376 Room-temperature X-ray diffraction (XRD) measurements were conducted using
377 a Bruker D8 Discover diffractometer with Cu K- α radiation ($\lambda = 1.5406 \text{ \AA}$) and
378 a 1 mm aperture collimator. Diffraction patterns were collected using a Vantec
379 500 area detector over a 2θ range of $30\text{--}80^\circ$. Phase identification and diffraction
380 pattern analysis were performed using the MAUD software package.

381 2.4. Characterization

382 The optimization targets five objectives that together span the quasi-static-to-
383 dynamic response space: (a) yield strength (YS), (b) the UTS/YS ratio (a proxy for
384 strain hardening capacity and resistance to strain localization), and (c) engineering
385 strain at UTS (ductility reserve before peak load), all three extracted from uniaxial
386 tensile tests; (d) the dynamic-to-quasi-static hardness ratio ($H_{\text{dyn}}/H_{\text{qs}}$), obtained
387 from nanoindentation and SHPB measurements, which quantifies how much the
388 alloy strengthens under impact loading; and (e) depth of penetration (DoP), com-
389 puted from finite-element ballistic simulations calibrated to the tensile and SHPB
390 data acquired within the same iteration—the only objective derived from simula-
391 tion rather than direct measurement. Campaign 1 used hardness as an objective;
392 here we replace it with the tensile-derived metrics, which connect more directly to
393 bulk deformation behavior and are less sensitive to local microstructural hetero-
394 geneity.

395 2.4.1. Quasistatic Tensile Deformation

396 Uniaxial tensile tests were conducted at ambient temperature using an MTS 810
397 servo-hydraulic testing system equipped with an 11.12 kN (2500 lbf) load cell, un-
398 der a constant quasi-static strain rate of $5 \times 10^{-4} \text{ s}^{-1}$. Axial strain was measured
399 using an MTS clip-on extensometer mounted directly on the specimen gauge sec-
400 tion. An initial force-controlled loading sequence was performed to verify speci-
401 men alignment, modulus consistency, and linearity of the elastic response. At least
402 2 tests were conducted for each condition to ensure reproducibility. Tensile tests
403 were terminated upon complete fracture. True stress-strain (σ - ϵ) curves were cal-
404 culated, from which the ultimate tensile strength (UTS), 0.2% offset yield strength
405 (YS), and uniform elongation (ϵ) were determined.

406 2.4.2. Nanoindentation

407 Each sample was indented using a KLA Instruments iMicro nanoindenter fitted
408 with a Berkovich diamond tip. Hardness (H) and reduced modulus (E_r) were de-
409 termined following the Oliver and Pharr method [47]. Conventional nanoindenta-
410 tion hardness values were obtained from strain-rate jump tests using a constant $\dot{P}/P$

protocol, with hardness reported at an indentation depth of 2000 nm. A minimum of 10 strain-rate jump indents was performed for each alloy. Indents were typically spaced 100 μm apart to prevent overlap between neighboring plastic zones. Strain-rate jump tests were conducted to assess rate-dependent mechanical behavior across the candidate alloys. These measurements were used to estimate indentation strain-rate sensitivity. The strain-rate sensitivities obtained from nanoindentation were cross-validated against SHPB results and were found to be in close agreement, typically within one standard deviation. Projected contact areas were corrected for pile-up for each individual indent. The projected contact area was calculated from the calibrated tip area function, $A(h_c)$, determined from fused silica calibration [47], and evaluated at the contact depth, h_c . The contact depth was calculated from the measured indentation depth, h , using an experimentally measured pile-up ratio,

$$\eta = \frac{h_c}{h}$$

The pile-up ratio was measured after each indentation using optical profilometry. First, the nominal indentation area, A_N , was determined from the region below the original sample surface. This area scales with h^2 . The actual residual contact area, A_R , was then segmented using a gradient-based method that identifies the Berkovich impression boundary; this area scales with h_c^2 . Assuming self-similar Berkovich geometry, the pile-up ratio was calculated as

$$\eta = \frac{h_c}{h} \approx \sqrt{\frac{A_R}{A_N}}$$

An independent value of η was measured for every indent and applied to the corresponding hardness and modulus calculation. This correction was applied to strain-rate jump, quasistatic, and dynamic hardness measurements. The pile-up correction typically reduced the measured hardness by approximately 5 to 15%, depending on alloy composition and testing condition. Custom analysis scripts were developed to extract the instrument data and automate projected contact area quantification; additional details of the image-analysis workflow are provided in the SI.

2.4.3. High Strain Rate Nanoindentation

High strain rate nanoindentation was performed on a separate custom instrument adapted for indentation impact testing at strain rates exceeding 10^3 s^{-1} . The instru-

441 ment includes a hexapod specimen stage and an InForce 1000 actuator customized
442 to measure displacements at high acquisition rates using a Michelson interferom-
443 eter; full details of this setup are provided by Hackett *et al.* (2023). All tests were
444 performed with a Berkovich three-sided pyramidal diamond tip.

445 The high strain rate instrument was used to measure the dynamic-to-quasistatic
446 hardness ratio, $H_{\text{DYN}}/H_{\text{QS}}$, which was incorporated into the Bayesian optimization
447 loop. Dynamic hardness, H_{DYN} , was measured from indentation impact tests with
448 an effective $\dot{P}/P = 2 \times 10^3 \text{ s}^{-1}$. Quasistatic hardness, H_{QS} , was measured on
449 the same instrument using a constant $\dot{P}/P = 2 \times 10^{-1} \text{ s}^{-1}$. A minimum of 10
450 dynamic impact indents and 10 quasistatic indents was performed for each alloy,
451 with indents typically spaced $100 \mu\text{m}$ apart.

452 Dynamic tests used an impact force of 20 mN, followed by reloading at the same
453 indentation location to 1 N at $\dot{P}/P = 2 \times 10^{-1} \text{ s}^{-1}$. The reloading step was in-
454 cluded to compare the dynamic-to-quasistatic hardness ratio obtained from sepa-
455 rate tests with the ratio obtained from a single indentation location; those results
456 will be presented in a separate study. Independent optical profilometry measure-
457 ments were used to determine η for each dynamic and quasistatic indent, and the
458 corresponding pile-up correction was applied to each hardness value before calcu-
459 lating $H_{\text{DYN}}/H_{\text{QS}}$.

460 Custom analysis scripts were developed to extract data from the instrument’s en-
461 crypted binary output and automated projected contact area quantification (see **SI**
462 for details). High strain rate hardness data was acquired for all candidate alloy
463 compositions and incorporated into the Bayesian optimization loop.

464 2.4.4. Split-Hopkinson Pressure Bar (SHPB) Testing

465 To characterize the high strain rate mechanical behavior of the high-entropy alloys
466 (HEAs) developed in this program, Split-Hopkinson Pressure Bar (SHPB) test-
467 ing was performed on all candidate alloys. For each composition, a minimum of
468 two specimens were tested, and approximately half of the alloys were evaluated at
469 multiple strain rates, namely 1500 s^{-1} or both 1500 s^{-1} and 3000 s^{-1} .

470 The SHPB experiments used a pulse-shaping technique to achieve and maintain a
471 uniform strain rate throughout the major portion of the specimen deformation. A
472 Ni-30Mo alloy was used as the pulse shaper due to its strain hardening behavior,
473 which closely resembles that of the tested HEAs. The SHPB setup captures three
474 essential waveforms: the incident pulse, transmitted pulse, and reflected pulse.

These signals are time-synchronized to construct the dynamic stress–strain response of the material. Figure 6 shows representative results from two SHPB tests performed at a nominal strain rate of 1500 s^{-1} . The black curves represent the true stress–strain response, while the blue curves illustrate the instantaneous strain rate as a function of strain. These measurements provide the dynamic response data required for the multi-objective optimization framework.

2.5. Depth of Penetration Simulation

Viscoplasticity Constitutive Model Implementation: We modeled the ballistic impact response using the Cowper-Symonds (CS) viscoplasticity framework to capture strain rate sensitivity under dynamic loading. The CS model provides a description of the flow stress evolution as a function of plastic strain rate through a multiplicative enhancement factor applied to the quasi-static flow strength. The constitutive relationship is expressed as:

$$\sigma_f = \sigma_{qs}^f \left(1 + \left(\frac{\dot{\epsilon}_p}{D} \right)^{1/M} \right) \quad (1)$$

where σ_{qs}^f represents the quasi-static flow strength that evolves with accumulated plastic strain due to work hardening mechanisms, $\dot{\epsilon}_p$ denotes the effective plastic strain rate, $D = 10^6 \text{ s}^{-1}$ serves as a material-dependent constant multiplier, and M constitutes the strain rate sensitivity exponent within the CS formulation. This formulation captures dynamic strengthening at elevated loading rates, where dislocation motion becomes increasingly constrained.

In the Cowper–Symonds form, D and M are mathematically coupled: for a given pair of quasi-static and dynamic flow-stress measurements, a continuum of (D, M) combinations can reproduce the data with comparable fidelity. With one SHPB curve and one quasi-static curve per alloy, joint identification of both parameters is underdetermined. We resolve this by fixing $D = 10^6 \text{ s}^{-1}$ uniformly across all alloys and calibrating M per alloy by RMSE minimization against the experimental QS+SHPB data (Eq. 2). This convention isolates the alloy-to-alloy variation in a single rate-sensitivity parameter, allows direct comparison of M across the composition set on a common basis.

Rate Sensitivity Parameter Identification: The strain rate sensitivity parameter M for each high entropy alloy composition was determined through a systematic calibration procedure based on experimental data obtained from quasistatic uniax-

506 ial tension tests and dynamic SHPB compression experiments conducted at strain
 507 rates ranging from 10^{-3} s^{-1} to 10^4 s^{-1} . The parameter identification process used a
 508 nonlinear least-squares optimization algorithm to minimize the root mean squared
 509 error (RMSE) between the CS model predictions and experimental true stress mea-
 510 surements:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (\sigma_{pred,i}^f - \sigma_{obs,i}^f)^2} \quad (2)$$

511 where N represents the total number of discrete data points in the experimental
 512 stress-strain response, $\sigma_{pred,i}^f$ denotes the model-predicted flow stress at the i -th
 513 data point, and $\sigma_{obs,i}^f$ corresponds to the experimentally observed true stress value.
 514 The optimization procedure was implemented under isothermal conditions to iso-
 515 late the strain rate effects from thermal softening mechanisms. The calibrated rate
 516 sensitivity exponents for the investigated high entropy alloy compositions ranged
 517 between 2 and 7, consistent with the expected strain rate sensitivity behavior of
 518 face-centered cubic polycrystalline materials.

519 *Finite Element Ballistic Impact Simulation:* We implemented the calibrated mate-
 520 rial parameters into a user-defined material subroutine (UMAT) within the ABAQUS/Explicit
 521 finite element framework to simulate the ballistic impact response. The compu-
 522 tational model used a two-dimensional axisymmetric formulation to simulate the
 523 normal impact of a rigid spherical projectile onto a cylindrical target of the can-
 524 didate alloy. The target geometry consisted of a cylinder with radius $r_t = 34 \text{ mm}$
 525 and thickness $t = 17 \text{ mm}$, while the projectile was modeled as a rigid sphere with
 526 diameter $d_p = 10 \text{ mm}$ impacting at a velocity of $V_0 = 500 \text{ m/s}$.

527 The finite element discretization used explicit, linear, axisymmetric stress elements
 528 (CAX4R) with a characteristic mesh size of 0.8 mm , resulting in a computational
 529 domain of 42×21 elements in the radial and axial directions, respectively. The tar-
 530 get material properties included an elastic modulus $E_t = 200 \text{ GPa}$, Poisson's ratio
 531 $\nu_t = 0.3$, and density $\rho_t = 7.85 \times 10^{-9} \text{ tonne/mm}^3$ (ABAQUS mm–tonne–s–N con-
 532 sistent units). Mesh distortion controls were implemented using default ABAQUS
 533 settings. The elastic constants and density are held fixed across all the alloys. This
 534 reflects both the available characterization data and a deliberate scoping choice.
 535 The quasi-static tension and SHPB experiments resolve only the plastic flow re-
 536 sponse of each alloy; elastic moduli and densities were not independently mea-

537 sured for individual compositions, so representative steel-class values ($E_t = 200$
538 GPa, $\rho_t = 7.85 \times 10^{-9}$ tonne/mm³) were used uniformly. For Ni-stabilized FCC
539 HEAs representative of the type studied here, Young’s modulus and Poisson’s ratio
540 are expected to vary by less than $\sim 10\%$ across compositions and to contribute neg-
541 ligibly to penetration response, which is dominated by plastic dissipation at 500
542 m/s.

543 *Depth of Penetration Quantification:* The ballistic performance of each alloy com-
544 position was quantified through the depth of penetration (DoP), defined as the
545 maximum axial displacement of the projectile centroid into the target at the in-
546 stant when the projectile velocity approaches zero. DoP serves as the primary re-
547 sponse variable for correlating material properties with impact performance. All
548 alloys across the three optimization iterations were evaluated under identical load-
549 ing conditions using their individually calibrated viscoplastic constitutive models.

550 2.6. Computational Design Framework

551 The computational component of BRAVE linked thermodynamic screening, physics-
552 informed surrogate modeling, feasibility classification, multi-objective acquisi-
553 tion, and diversity-aware batch selection into one loop. In each iteration, exper-
554 imentally verified FCC alloys provided objective data for surrogate updates, while
555 infeasible alloys contributed only their phase labels to the feasibility model. A
556 more detailed end-to-end workflow diagram is provided in the SI.

557 2.6.1. Surrogate Models and Objective Representation

558 We used Gaussian process surrogate models because they provide both property
559 predictions and uncertainty estimates, which are needed for Bayesian optimization.
560 The surrogates were initialized with informative priors rather than zero-mean pri-
561 ors, using the Varvenne–Luque–Curtin solid solution strengthening model together
562 with prior campaign data to provide a physics-grounded starting point before sub-
563 stantial new data were collected. Detailed mathematical expressions for the GP
564 posterior and prior construction are provided in the Appendix.

565 The optimization targeted the five objectives defined in Section 2.3. Four objec-
566 tives (YS, UTS/YS, strain at UTS, and $H_{\text{dyn}}/H_{\text{qs}}$) were obtained experimentally,
567 while DoP was computed from constitutive calibration and FEM simulation using
568 data acquired in the same iteration. Because EHVI is implemented in minimization
569 form, the four desirable-to-maximize objectives were sign-inverted before surro-
570 gate modeling and hypervolume calculation, while DoP remained in its original
571 minimization form.

2.6.2. Acquisition, Fusion, and Batch Selection

Candidate ranking was based on Expected Hypervolume Improvement (EHVI), which favors alloys expected to improve the current Pareto front while still exploring uncertain regions. Where multiple computational estimates contributed to the same target, their predictions were fused through a reification-based multi-source model so that the acquisition function operated on a single posterior distribution for each objective. To reduce sensitivity to any one kernel hyperparameter setting in the low-data regime, we used an ensemble of Gaussian processes with different length-scale assumptions and then applied k -medoids clustering to select a diverse experimental batch. Mathematical details for EHVI, reification, and the batch ensemble strategy are provided in the Appendix.

2.6.3. Constraint-Aware Acquisition with Learned Feasibility

Phase stability constraints were enforced initially through CALPHAD screening, but deterministic rejection alone is too rigid near the FCC phase boundary. We therefore trained a Gaussian process classifier (GPC) on experimentally verified phase outcomes to estimate a continuous feasibility probability, $P_{\text{feas}}(\mathbf{x})$, for each candidate composition. This allowed the optimizer to distinguish compositions that were confidently infeasible from those that were merely uncertain.

To assess how robust CALPHAD-predicted FCC stability is to small local composition changes, we performed a perturbation analysis on candidate alloys that satisfied $\text{FCC} \geq 0.99$. Of the 1,154,207 perturbed compositions evaluated, 903,326 (78.3%) retained $\text{FCC} \geq 0.99$. We therefore interpreted a feasibility probability near 0.8 as corresponding to a composition that is locally robust to small perturbations, which motivated the acquisition threshold $P_{\text{feas}}(\mathbf{x}) \geq 0.8$. Details are provided in the SI.

Candidates that passed the initial CALPHAD filter were assigned $P_{\text{feas}} = 1$. Candidates rejected by CALPHAD but predicted by the GPC to satisfy $P_{\text{feas}}(\mathbf{x}) \geq 0.8$ were reintroduced into the search space, but their acquisition value was penalized according to feasibility risk:

$$\text{EHVI}(\mathbf{x})_{\text{constraint-aware}} = \text{EHVI}(\mathbf{x}) \times P_{\text{feas}}(\mathbf{x}) \quad (3)$$

This feasibility-weighted acquisition suppressed high-risk candidates whose apparent promise was driven mainly by uncertainty, while still allowing near-boundary alloys to be selected when their predicted performance was sufficiently strong. As new alloys were synthesized and phase-verified, the GPC was retrained so that the

feasibility boundary improved over successive iterations. Manufacturing failures were not modeled by this classifier; it addressed phase-outcome feasibility only.

2.6.4. Summary of the BRAVE Campaign Algorithm

Figure 7 illustrates the data flow within a single iteration of the BRAVE campaign. The bifurcation at phase verification was the defining structural feature: feasible alloys proceeded through full characterization and fed objective data into the GP surrogates, while infeasible alloys contributed only their phase outcome to the GPC feasibility classifier. Both paths converged at the acquisition step, where EHVI was weighted by the learned feasibility probability. Algorithm 1 formalizes this workflow.

Algorithm 1 consolidates the full BRAVE campaign workflow, from design space initialization through iterative closed-loop optimization. Infeasible alloys (line 9) contribute to the feasibility classifier but not to surrogate model training, capturing the asymmetry between failed and successful experiments that motivates the risk-aware acquisition strategy. The DoP objective (lines 16–18) is computed within each iteration from experimentally calibrated constitutive models, creating the simulation-in-the-loop architecture described in Section 1.

The campaign ran for three iterations (48 alloys total). This budget was set by experimental throughput—vacuum arc melting, thermomechanical processing, and multi-technique characterization limit the number of alloys that can be completed per cycle—rather than by a convergence criterion.

2.7. Data Management and Overall Workflow

Data management in this study follows a method-centric architecture with IGSN (International Generic Sample Number) registration for all physical samples. Details of the data architecture, folder structure, IGSN metadata schema, FAIR compliance, and a more detailed end-to-end workflow diagram are provided in the SI.

3. Results and Discussion

3.1. Phase Verification and Feasibility Outcomes

The 16 candidate alloys selected for iteration BBA were chosen by diversity-aware k-medoids sampling across the 37 chemical subsystems (Section 2.1.3). Figure 8 shows the target chemistries of all 48 alloys synthesized in this work, projected

Algorithm 1 BRAVE: Bayesian Risk-Aware Alloy Discovery and Exploration

Input: 8-element composition grid, CALPHAD database, batch size $B = 16$

// Initialization

- 1: Apply compositional bounds $\rightarrow$ 1,504,938 candidates
- 2: CALPHAD screen (TCHEA6, FCC ≥ 0.99 at 700°C) $\rightarrow$ 26,621 feasible
- 3: Partition feasible space into 37 chemical subsystems
- 4: Retain subsystem labels for feasible candidates
- 5: k -medoids clustering $\rightarrow$ 1,000 representative alloys
- 6: Diversity-aware down-selection $\rightarrow B_0 = 16$ initial candidates

// Mid-campaign update (from iteration BBB onward)

- 7: Augment feasibility screening using TCHEA7 (700°C and 950°C)
- 8: Update feasible design space

// Iterative closed-loop optimization

- 9: **for** iteration $t = 0, 1, 2$ **do**
- 10: Synthesize B_t alloys via vacuum arc melting
- 11: Verify phase purity (XRD): classify each alloy as *feasible* or *infeasible*
- 12: Update GPC feasibility classifier with new phase outcomes
- 13: **for** each *feasible* alloy **do**
- 14: Characterize: tensile $\rightarrow$ YS, UTS/YS, strain at UTS
- 15: Characterize: NI + HSRNI + SHPB $\rightarrow H_{\text{dyn}}/H_{\text{qs}}$
- 16: Calibrate Cowper–Symonds model from tensile + SHPB data
- 17: Simulate depth of penetration (DoP) via FEM with calibrated model
- 18: **end for**
- 19: Update GP surrogates with new objective measurements (feasible alloys only)
- 20: Fuse multi-source predictions via reification
- 21: **for** each of n random length-scale configurations **do**
- 22: Build GP ensemble member
- 23: Compute $\text{EHVI}(\mathbf{x}) \times P_{\text{feas}}(\mathbf{x})$ for all remaining candidates
- 24: **end for**
- 25: k -medoids select B_{t+1} from top-ranked candidates
- 26: **end for**

Output: Pareto-optimal alloys, trained surrogates, feasibility classifier

636 from the 8-element composition simplex onto a regular octagon by affine coordi-
637 nates and colored by iteration. This view emphasizes how the campaign progres-
638 sively populated distinct regions of the accessible composition space across BBA,
639 BBB, and BBC. A more detailed visualization of the iteration-0 batch selection
640 is provided in the SI. All candidates across the three iterations were synthesized
641 via vacuum arc melting, cold rolled, and recrystallized as described in Section 2.2.
642 Phase purity was assessed by XRD for every alloy prior to mechanical testing.

643 Figure 9(a–c) presents the diffraction patterns across all three iterations: the ma-
644 jority of alloys show single-phase FCC reflections at (111), (200), and (220), con-
645 sistent with the CALPHAD-based design targets. Secondary intermetallic sigma
646 ($P4_2/mnm$) phase was detected in a subset of alloys from iterations BBB and BBC.

647 Sigma phase appeared preferentially in alloys with combined V + Cr content $\geq$
648 20 at.%, although several alloys with high V + Cr (BBA09, BBB01, BBB03,
649 BBC02, BBC04) remained single-phase FCC. These exceptions had Ni content of
650 36 at.% or higher, consistent with increased valence electron concentration (VEC)
651 destabilizing the sigma phase. We note that this compositional dependence—
652 sigma formation correlating with V + Cr and suppressed by Ni—directly informed
653 the GPC feasibility classifier training data.

654 The sigma-phase failures in BBB triggered a transition from the TCHEA6 to TCHEA7
655 database and the addition of a 950°C FCC stability constraint (Section 2.1.2).
656 Comparing TCHEA7 predictions with experimental XRD results for iterations
657 BBA and BBB revealed five false positives (secondary phases predicted but not
658 observed) and two false negatives (FCC predicted but sigma observed) at 700°C,
659 and one false positive and four false negatives at 950°C. False negatives are the
660 more consequential error: they lead to failed experiments (i.e., budget spent with-
661 out usable objective data). These prediction discrepancies quantify the CALPHAD
662 model uncertainty that the GPC feasibility classifier is designed to absorb.

663 Figure 10(a–c) presents BSE-SEM images of three representative alloy composi-
664 tions (one per iteration), showing fully recrystallized, equiaxed FCC grain struc-
665 tures with annealing twins. No significant porosity or macrosegregation was ob-
666 served. The corresponding EBSD inverse pole figure (IPF) maps (Figure 10(d–f))
667 confirm the absence of textural anisotropy, indicating that recrystallization reset
668 the deformation-induced orientation bias from cold rolling. Grain size distribu-
669 tions (Figure 10(g–i)) show average grain sizes between 10 μm and 35 μm , with
670 compositionally complex alloys (7–8 elements) showing finer grains and narrower
671 distributions, consistent with sluggish diffusion in concentrated solid solutions.

Table 1. Composition and phase analysis of BBA (Iteration 0) alloys

Alloy Name	Target Atomic %								EDS Measured Atomic %								XRD Phase
	Al	Co	Cr	Cu	Fe	Mn	Ni	V	Al	Co	Cr	Cu	Fe	Mn	Ni	V	
BBA01	4	8	4	4	16	12	48	4	4.1	8.2	4.2	4.2	16.1	12.2	47.0	4.1	FCC
BBA02	4	16	0	4	12	8	52	4	4.2	15.0	0.0	4.2	12.2	8.2	52.2	4.1	FCC
BBA03	4	12	8	4	16	8	48	0	4.3	12.1	8.2	4.2	16.1	8.2	47.0	0.0	FCC
BBA04	0	12	8	4	16	20	36	4	.	.	.	.	.	.	.	.	—
BBA05	4	12	8	0	8	8	56	4	4.1	12.2	8.2	0.0	8.2	8.0	55.2	4.1	FCC
BBA06	4	0	8	4	24	12	44	4	4.2	0.0	8.3	4.1	24.0	11.9	43.4	4.1	FCC
BBA07	4	12	0	0	16	8	52	8	5.0	12.3	0.0	0.0	16.2	7.9	50.5	8.1	FCC
BBA08	0	24	8	0	12	16	32	8	0.0	22.9	8.1	0.0	12.2	16.0	32.7	8.1	FCC
BBA09	4	16	8	0	12	0	48	12	4.9	16.1	8.2	0.0	12.0	0.0	46.6	12.1	FCC
BBA10	4	0	4	0	20	8	52	12	4.4	0.0	4.2	0.0	20.2	8.0	51.0	12.2	FCC
BBA11	0	16	0	4	20	20	28	12	0.0	16.1	0.0	4.0	20.1	20.3	27.4	12.0	FCC
BBA12	0	0	8	4	16	12	52	8	0.0	0.0	8.3	4.1	16.1	12.3	51.2	8.0	FCC
BBA13	4	20	4	4	16	0	52	0	4.0	20.4	4.1	4.2	16.3	0.0	51.0	0.0	FCC
BBA14	0	16	8	4	16	20	36	0	0.0	16.1	8.2	4.0	16.0	20.3	35.3	0.0	FCC
BBA15	0	20	8	4	0	20	44	4	0.0	19.9	8.2	4.2	0.0	20.5	43.2	4.0	FCC
BBA16	4	12	0	12	20	8	44	0	4.1	12.3	0.0	11.8	20.4	8.1	43.2	0.0	FCC

Alloys confirmed as single-phase FCC proceeded to full mechanical characterization, and we used their objective measurements to train the GP surrogates. Alloys containing sigma phase were classified as infeasible for the purposes of Bayesian optimization: their phase outcome was used to retrain the GPC feasibility classifier, but their objective measurements (when obtainable) were not included in surrogate model training. Several sigma-bearing alloys (BBC03, BBB13, BBC08, BBC14) retained sufficient FCC matrix to yield tensile specimens and were mechanically tested for scientific characterization, but their properties were excluded from the BO loop. Alloys that failed during processing (BBC05–07, BBC09–10, BBC16, BBA04) produced no specimens and contributed only their infeasible label to the GPC. Table 1, Table 2, and Table 3 list the target and measured compositions (EDS) alongside XRD phase verification for all three iterations. Candidates that violated project constraints (manufacturing failures or secondary phases) are highlighted in red.

3.2. Mechanical Properties Across Iterations

3.2.1. Tensile Response

Figure 11 presents the true stress–strain curves for all tested FCC alloys across the three iterations. True fracture stress increased progressively across successive iterations, with the majority of alloys surpassing 1 GPa in true tensile strength

Table 2. Composition and phase analysis of BBB (Iteration 1) alloys

Alloy Name	Target Atomic %								EDS Measured Atomic %								XRD Phase
	Al	Co	Cr	Cu	Fe	Mn	Ni	V	Al	Co	Cr	Cu	Fe	Mn	Ni	V	
BBB01	0	24	4	0	8	4	36	24	0.0	20.5	4.2	0.0	8.1	4.0	38.9	24.4	FCC
BBB02	4	32	8	4	4	20	24	4	4.2	30.4	8.2	4.1	4.1	20.1	24.7	4.1	FCC
BBB03	4	0	12	4	4	16	52	8	4.1	·	12.3	4.1	4.1	16.1	51.2	8.2	FCC
BBB04	4	12	4	0	28	0	40	12	4.0	11.1	4.1	0.0	28.0	0.0	40.8	12.0	FCC
BBB05	0	44	12	0	4	4	12	24	·	·	·	·	·	·	·	·	FCC + σ
BBB06	0	16	20	0	4	4	36	20	·	·	·	·	·	·	·	·	FCC + σ
BBB07	4	40	0	0	12	4	36	4	4.0	39.1	0.0	0.0	12.3	3.9	36.7	4.0	FCC
BBB08	4	8	12	4	4	24	40	4	3.8	8.0	12.3	4.2	4.1	24.0	39.4	4.1	FCC
BBB09	0	4	8	0	4	28	40	16	·	·	·	·	·	·	·	·	FCC + σ
BBB10	4	0	4	4	8	16	52	12	·	·	·	·	·	·	·	·	FCC
BBB11	0	24	4	0	4	32	20	16	·	·	·	·	·	·	·	·	FCC + σ
BBB12	0	20	4	0	4	4	48	20	0.0	20.2	4.2	0.0	4.1	4.0	47.1	20.4	FCC
BBB13	0	40	4	0	28	4	4	20	0.0	39.7	4.2	0.0	27.9	3.9	4.2	20.1	FCC + σ
BBB14	4	0	12	16	4	12	52	0	4.4	0.0	12.1	15.7	4.1	12.4	51.3	0.0	FCC
BBB15	4	0	8	24	4	16	44	0	4.3	0.0	8.3	23.5	4.1	16.2	43.7	0.0	FCC
BBB16	4	36	0	0	4	4	40	12	·	·	·	·	·	·	·	·	FCC

by iteration BBB. Ductility remained comparable across iterations, with the top-performing alloys sustaining strain at UTS in the range of 45–50%. Enhanced strain hardening across iterations delayed the onset of strain localization, manifesting as high UTS/YS ratios. Compositions with UTS/YS > 3.5 demonstrated superior strain hardening capacity, contributing to delayed failure under quasi-static loading. For context, Campaign 1 achieved UTS/YS ratios ranging from 1.38 to 2.93 [46]. In Campaign 2, every successfully tested single-phase FCC alloy exceeded Campaign 1’s maximum: the minimum UTS/YS was 3.07 (BBA07), the median was 3.79, and 9 alloys exceeded 4.0. We note that this wholesale improvement over Campaign 1 is not attributable to the optimizer alone. The GP surrogates in Campaign 2 were initialized with informative priors trained on Campaign 1 experimental data and the Varvenne–Luque–Curtin solid solution strengthening model (Section 2.6), giving the optimizer a physics-grounded starting point rather than a zero-mean prior. The design space was shaped by Campaign 1’s experience: we expanded from 6 to 8 elements based on the observation that the 6-element space lacked sufficient compositional freedom to simultaneously maximize strength and ductility, and we replaced hardness with tensile-derived objectives (YS, UTS/YS, strain at UTS) based on the finding that hardness alone is an incomplete proxy for bulk mechanical response. The UTS/YS shift from 1.38–2.93 (Campaign 1) to 3.07–4.53 (Campaign 2) thus reflects the compounding effect of

Table 3. Composition and phase analysis of BBC (Iteration 2) alloys

Alloy Name	Target Atomic %								EDS Measured Atomic %								XRD Phase
	Al	Co	Cr	Cu	Fe	Mn	Ni	V	Al	Co	Cr	Cu	Fe	Mn	Ni	V	
BBC01	4	12	4	4	20	16	32	8	4.0	12.2	4.2	4.0	20.1	16.0	31.4	8.2	FCC
BBC02	4	8	0	0	4	8	52	24	4.0	8.2	0.0	0.0	4.1	8.0	51.2	24.4	FCC
BBC03	4	4	12	0	0	4	52	24	4.0	4.1	12.2	0.0	0.0	3.9	51.3	24.4	FCC + σ
BBC04	4	0	4	0	4	4	60	24	4.0	0.0	4.2	0.0	4.1	4.0	59.3	24.5	FCC
BBC05	0	24	8	0	16	40	4	8	.	.	.	.	.	.	.	.	—
BBC06	4	28	4	0	4	4	44	12	.	.	.	.	.	.	.	.	—
BBC07	4	32	4	0	0	4	44	12	.	.	.	.	.	.	.	.	—
BBC08	0	8	0	4	4	28	36	20	.	8.1	.	4.1	4.2	28.1	35.7	19.9	FCC + σ
BBC09	0	36	4	0	16	16	8	20	.	.	.	.	.	.	.	.	—
BBC10	4	4	16	0	8	0	52	16	.	.	.	.	.	.	.	.	—
BBC11	4	12	4	4	12	16	36	12	4.3	12.2	4.2	3.9	12.2	15.9	35.1	12.3	FCC
BBC12	4	4	4	8	28	16	32	4	4.1	4.3	4.2	7.8	28.0	16.1	31.3	4.1	FCC
BBC13	4	28	0	20	4	8	36	0	4.3	28.0	.	19.9	4.1	8.4	35.4	.	FCC
BBC14	0	12	4	0	4	16	40	24	.	12.1	4.1	.	4.3	16.1	39.0	24.5	FCC + σ
BBC15	4	16	0	20	4	12	44	0	4.4	16.2	.	19.7	4.2	12.3	43.3	.	FCC
BBC16	4	28	4	0	8	0	40	16	.	.	.	.	.	.	.	.	—

711 prior-informed surrogates, a better-chosen design space, and more directly relevant
712 optimization objectives. Tabulated values of YS, UTS, and strain to failure for all
713 tested alloys are provided in the SI.

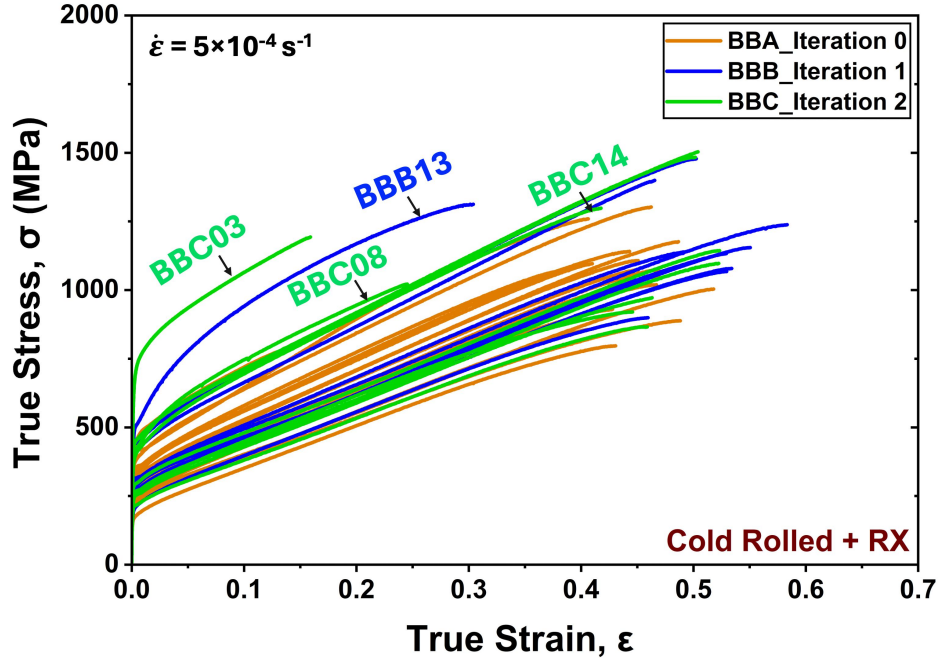


Figure 11. Consolidated true stress–strain curves from ambient temperature quasistatic uniaxial tensile tests for all tested FCC alloys across 3 iterations. Alloys containing sigma phase are indicated by black arrows. RX refers to recrystallized alloys.

3.2.2. Dynamic Hardness and Rate Sensitivity

Dynamic hardness consistently exceeded quasi-static hardness across all tested compositions (Figure 12), consistent with the higher strain rates associated with impact testing ($\sim 10^3 \text{ s}^{-1}$) relative to quasi-static indentation ($\sim 10^0 \text{ s}^{-1}$). Among single-phase FCC alloys, BBB12 showed the highest hardness. BBC03, which contained secondary sigma phase, showed the overall highest hardness.

The dynamic-to-quasi-static hardness ratio, $H_{\text{DYN}}/H_{\text{QS}}$ (Figure 13), ranged from 1.09 to 1.22 across all samples, with no clear compositional trend. Error bars were computed via propagation of uncertainty from the standard deviations of both dynamic and quasi-static measurements. None of the tested compositions showed error bars overlapping with unity, confirming a measurable rate-dependent strengthening effect in every alloy.

The contact-depth-to-indentation-depth ratio (h_c/h) was close to 1 for most samples, indicating minimal pile-up or sink-in. BBC03 was an exception (h_c/h up to

1.03), consistent with its sigma-phase-induced deformation behavior. The near-unity h_c/h for single-phase FCC alloys supports the use of nominal hardness calculations, reducing the need for the time-consuming imaging and tracing steps required for precise contact area estimation.

3.2.3. Depth of Penetration

Simulated DoP values across all iterations ranged from 2.22 to 3.35 mm (tabulated in **SI**). The five best-performing alloys (lowest DoP)—BBC03, BBB13, BBC02, BBB01, and BBC04—were concentrated in iterations BBB and BBC, while the five worst-performing alloys (BBA13, BBA16, BBC15, BBA03, and BBB07) were predominantly from iteration BBA, with one (BBC15) from iteration BBC. This shift confirms that the Bayesian optimization loop progressively identified compositions with improved impact resistance. The 1.13 mm spread in DoP across the full dataset (3.35 – 2.22 mm), though narrow in absolute terms, provided sufficient differentiation for the EHVI acquisition function to distinguish candidates.

3.2.4. Property Evolution

[Figure 14](#) summarizes the statistical evolution of all measured properties across the three iterations. Median values for strength-related properties (YS, UTS, H_{dyn} , H_{qs}) increased from BBA to BBC while inter-quartile ranges narrowed, indicating convergence toward consistently high-performing compositions. Median DoP decreased across iterations, with the distribution tightening around lower penetration depths. Strain at UTS was maintained or improved despite the strength gains, confirming that the multi-objective framework avoided the classical strength–ductility trade-off.

These improvements in median properties should be interpreted alongside a processing caveat. The optimization campaign treats composition as the sole design variable, but the measured properties also depend on thermomechanical processing history—specifically, the degree of cold work, recrystallization temperature, and resulting grain size. Several alloys that passed phase verification (single-phase FCC by XRD) showed anomalous tensile behavior: low ductility, premature failure, or high scatter between replicate specimens. Post-mortem analysis traced these anomalies to incomplete recrystallization or non-uniform grain size distributions rather than compositional effects. The consequence for the optimization loop is twofold. First, noisy property measurements reduce the signal-to-noise ratio available to the GP surrogates, slowing convergence. Second, alloys whose measured properties reflect processing defects rather than intrinsic compositional effects can mislead the acquisition function, directing the search toward or away

764 from regions for the wrong reasons. Grain sizes in this campaign ranged from 10
765 to 35 μm (Figure 10), and the more compositionally complex alloys (7–8 elements)
766 generally showed finer, more uniform grains—consistent with sluggish diffusion
767 reducing grain growth—and correspondingly lower property scatter. Standard-
768 izing the thermomechanical route to produce consistent, fully recrystallized mi-
769 crostructures across all compositions is a prerequisite for isolating compositional
770 effects in future campaigns.

771 3.3. Composition–Property Relationships

772 3.3.1. Correlation Structure and Feature Importance

773 Figure 15 combines three complementary analyses: (a) a Pearson correlation ma-
774 trix quantifying linear relationships among elemental concentrations and mechan-
775 ical responses, (b) a SHAP beeswarm plot showing each element’s importance and
776 directional behavior in the predictive model, and (c) a corrSHAP bar chart mea-
777 suring the correlation between each element’s compositional value and its SHAP
778 contribution. SHAP values were computed from the gradient-boosted regression
779 models trained independently on each of the five target objectives normalized per
780 target, and pooled to give a single multi-objective view. The multi-objective corr-
781 SHAP (Figure 15c) is the per-target Pearson correlation between feature value and
782 SHAP value, averaged over the five target models. Positive values mean the el-
783 ement systematically raises the composite response across targets; cancellation
784 across targets reflects a multi-objective trade-off.

785 Vanadium and copper define the primary compositional axis governing strength
786 and penetration resistance. Vanadium shows strong positive correlations with YS
787 ($r = 0.82$, $n = 37$; the $n = 37$ set includes all alloys with measurable YS regard-
788 less of feasibility outcome, so this figure is not inflated by feasibility filtering),
789 UTS ($r = 0.80$), and both hardness measures ($r \approx 0.81$), and a strong negative
790 correlation with DoP ($r = -0.88$). Copper shows the opposite pattern: negative
791 correlations with all strength metrics ($r = -0.44$ to -0.55) and positive correla-
792 tion with DoP ($r = 0.53$). The V–Cu anticorrelation ($r = -0.61$) means the design
793 space naturally partitions along this axis.

794 Among property–property relationships, YS and hardness are tightly coupled ($r =$
795 0.88), and the two hardness metrics are nearly identical ($r = 0.99$). UTS/YS is
796 strongly anticorrelated with YS ($r = -0.91$) and positively correlated with elonga-
797 tion ($r = 0.80$), confirming that strain-hardening capacity accompanies ductility.
798 DoP correlates almost perfectly (inversely) with YS ($r = -0.94$) and strongly with

799 UTS ($r = -0.86$), indicating that penetration resistance in this alloy family is gov-
800 erned predominantly by strength.

801 SHAP and corrSHAP reflect the chosen multi-objective target and not single-
802 property linear correlations. The SHAP results align with the linear correlations.
803 Vanadium dominates mean |SHAP|, with normalized contributions spanning the
804 full $[-1, +1]$ range; high-V points (red) at the positive tail and low-V (blue) at
805 the negative tail, matching V's positive correlations with strength metrics. Iron,
806 manganese, and nickel follow; aluminum has the smallest mean |SHAP| and clus-
807 ters near zero. Mn and Ni dots show mixed colors at both positive and negative
808 SHAP values, indicating interaction-mediated (non-monotonic) effects whose sign
809 depends on the concentrations of other elements.

810 Aluminum has the strongest positive corrSHAP ($r \approx +0.78$), significantly higher
811 than iron (+0.27), chromium (+0.22), and copper (+0.15) — higher Al consis-
812 tently pushes the composite up despite Al's modest |SHAP| magnitude. This is
813 consistent with aluminum's role as a solid solution strengthener in Ni-stabilized
814 HEAs, where small additions of Al improve YS and UTS/YS without strongly
815 penalizing ductility. Nickel is near zero ($r \approx 0$), confirming the sign-flipping,
816 interaction-mediated behavior seen in SHAP beeswarm plot. The negative tail
817 is vanadium (-0.32), cobalt (-0.22), and manganese (-0.17). Vanadium's nega-
818 tive multi-objective corrSHAP coexists with its strong positive correlations with
819 strength because the composite also includes elongation and UTS/YS, with which
820 V correlates -0.36 and -0.67 , respectively. The ductility penalty associated with
821 V (likely tied to its role in promoting sigma phase formation and reducing strain
822 hardening capacity) dominates the strength benefit when all five objectives are
823 weighted together.

824 3.3.2. *Strength–Ductility Trade-off*

825 Figure 16 shows how the BBA (iteration 0), BBB (iteration 1), and BBC (iter-
826 ation 2) alloys evolve through key strength–ductility property spaces, illustrating
827 both improvement in feasible FCC alloys and the trade-off imposed by sigma phase
828 formation.

829 In panel (a), most BBA (iteration 0) alloys occupy the lower-strength region, while
830 BBB (iteration 1) and BBC (iteration 2) extend further into the upper-right por-
831 tion of the plot. BBC04 reaches ~ 450 MPa YS at ~ 1480 MPa UTS, while BBC03
832 achieves ~ 720 MPa YS but at lower UTS (~ 1160 MPa), consistent with sigma-
833 phase embrittlement. In panel (b), several of the highest-ductility, highest-work-

834 hardening alloys across the campaign, including BBB03, BBB08, BBB15, BBC01,
835 and BBA06, cluster above 45% elongation with UTS/YS > 3.5 . In panel (c), the
836 population follows the expected negative trend between elongation and YS, but
837 BBC04 and BBC02 (~ 450 – 490 MPa YS and ~ 48 – 50% elongation) sit above it. In
838 panel (d), BBB (iteration 1) and BBC (iteration 2) increasingly populate the upper-
839 right quadrant, with BBC04 combining ~ 1480 MPa UTS with $\sim 50\%$ elongation.

840 The sigma-bearing alloys BBC03 and BBB13 serve as counterexamples: elevated
841 YS comes at steep ductility cost, reinforcing the importance of phase stability as
842 a design constraint. A detailed analysis of the compositional and microstructural
843 factors driving these improvements will be presented in a forthcoming publication.

844 3.3.3. Strength–DoP Correlations and Rate Sensitivity

845 DoP shows the expected inverse correlation with strength metrics (Figure 17):
846 higher YS and UTS correspond to lower penetration depth. The strong DoP–YS
847 correlation ($r = -0.94$) raises the question of what DoP adds beyond YS alone.
848 The residual variance captures the contribution of rate-dependent strengthening
849 through the Cowper–Symonds constitutive model: alloys with the same YS but
850 different rate-sensitivity exponents m produce different DoP values, providing a
851 simulation-derived discriminant that quasi-static testing alone cannot.

852 The relationship between DoP and the calibrated rate-sensitivity exponent m (Fig-
853 ure 18(a)) is moderately positive, meaning that higher m values are associated with
854 larger penetration depths. YS and UTS both correlate negatively with m (Fig-
855 ure 18(b,c)), with the YS relationship stronger than the UTS relationship. Together,
856 these trends indicate that alloys with higher quasi-static strength tend to show lower
857 calibrated rate-sensitivity exponents in this dataset, consistent with a trade-off be-
858 tween quasi-static strength and rate sensitivity in the present Cowper–Symonds
859 fits.

860 3.3.4. Emergent Compositional Design Rules

861 The optimization campaign revealed two distinct compositional routes to high per-
862 formance within the FCC-feasible space, which we describe below.

863 The first is a *high-strength regime* defined by $V = 20$ – 24 at.%, $Cr \leq 4$ at.%, Ni
864 $= 36$ – 60 at.%, and $Cu = 0$. The four strongest single-phase FCC alloys—BBC04
865 (~ 1480 MPa UTS, 50% elongation), BBC02 (~ 1460 MPa, 48%), BBB01 (~ 1450 MPa,
866 49%), and BBB12 (~ 1400 MPa, 47%)—all share this compositional signature.
867 The strong V–YS correlation ($r = 0.82$, $n = 37$; Section 3.3.1) is consistent with

lattice distortion strengthening from the large atomic-size and modulus mismatch of V in the FCC matrix [48]. We note, however, that the Varvenne–Luque–Curtin model assumes a fully random solid solution; at V = 20–24 at.%, deviations from randomness (short-range ordering or local V clustering) could affect quantitative agreement, and this has not been independently validated for the alloys in this campaign. Nickel stabilizes the FCC phase at high V concentrations (VEC effect), and chromium is minimized to avoid sigma formation. BBC04 is the most extreme example: 60 at.% Ni and 24 at.% V with no Co, no Cu, and only 4 at.% Cr—an alloy at the outer edge of the design space that the BO identified precisely because the constraint-aware acquisition permitted boundary exploration.

The second is a *high-ductility regime* with moderate V (8–12 at.%), higher Mn (16 at.%), and moderate Cr. BBB03 (~1220 MPa UTS, 56% elongation, UTS/YS = 3.97) and BBC01 (~1070 MPa, 52%, UTS/YS = 4.20) represent this compositional region. The elevated UTS/YS ratios and sustained elongation in these Mn-rich alloys are consistent with enhanced strain hardening, possibly through SFE reduction and twinning [9, 10], although direct microstructural evidence of the operating deformation mechanisms was not obtained in this campaign. These alloys trade peak strength for superior ductility and strain hardening capacity.

The two routes occupy different regions of the Pareto front and validate the 8-element expansion from Campaign 1. The high-strength route exploits V–Ni synergy that was inaccessible in the 6-element system (which lacked the compositional freedom to simultaneously maximize V and Ni while suppressing Cr). The high-ductility route exploits Mn, which was absent from Campaign 1 entirely. Cu, by contrast, showed negative Pearson correlations with all strength metrics ($r = -0.44$ to -0.51 ; Section 3.3.1) while its multi-objective corrSHAP is slightly positive ($r \approx +0.15$), reflecting partial cancellation across the five targets; it appeared in none of the top-performing alloys. The BO effectively learned to deprioritize Cu-containing compositions—a result that informs future campaign design in this alloy family.

3.4. Bayesian Optimization Performance and Feasibility Analysis

Sections 3.1–3.3 reported the properties and composition–property relationships across the three iterations. The optimization metrics below quantify how the BRAVE framework produced these outcomes, and what the pattern of successes and failures exposes about the structure of the constrained design problem.

Table 4 summarizes the optimization performance across the three iterations. The

Table 4. Bayesian Optimization Performance Across Iterations

Iteration	Avg EHVI	Hypervolume	HV Gain	Pareto Size	Infeasible
BBA	N/A	658.24	—	10	1/16
BBB	836.41	817.77	24.2%	17 (9 new)	5/16
BBC	1059.2	856.30	4.7%	21 (5 new)	9/16

903 initial Pareto front (BBA) contained 10 non-dominated alloys with a hypervol-
 904 ume of 658.24. Iteration BBB increased the hypervolume to 817.77 (24.2% gain)
 905 and expanded the Pareto front to 17 alloys, with 9 new non-dominated composi-
 906 tions. Iteration BBC reached a hypervolume of 856.30 (4.7% gain) with 21 Pareto-
 907 optimal alloys.

908 The Bayesian advantage of the campaign is best measured by the concentration
 909 of experimental budget in high-value regions of the design space, rather than by
 910 Pareto front size or hypervolume trajectory—both of which are expected to in-
 911 crease under random sampling in correlated objective spaces. The campaign ex-
 912 plored 0.12% of the feasible space (33 feasible alloys out of 27,240 candidates)
 913 and concentrated four of its strongest alloys (BBC04, BBC02, BBB01, BBB12—
 914 all with $V \geq 20$ at.% and $Ni \geq 36$ at.%) in a compositional region comprising ap-
 915 proximately 100 of the 27,240 feasible candidates ($<0.4\%$ of the feasible space).
 916 Under random sampling, the probability of placing four or more of 33 draws in this
 917 region is $P \approx 6.5 \times 10^{-6}$ (hypergeometric distribution). The feasibility-weighted
 918 acquisition strategy is further validated by a controlled computational benchmark
 919 (Appendix), in which risk-weighted EHVI matched oracle performance (perfect
 920 feasibility knowledge) while reducing wasted evaluations approximately fivefold
 921 relative to feasibility-blind BO, across two bi-objective optimization problems on
 922 the 6-component Al–V–Cr–Fe–Co–Ni system.

923 3.4.1. Feasibility–Performance Trade-off

924 The number of alloys infeasible for Bayesian optimization (sigma phase or man-
 925 ufacturing failure preventing XRD verification) increased across iterations: 1/16
 926 in BBA (BBA04, manufacturing failure), 5/16 in BBB (BBB05, BBB06, BBB09,
 927 BBB11, BBB13—all sigma), and 9/16 in BBC (BBC03, BBC08, BBC14 sigma;
 928 BBC05, BBC06, BBC07, BBC09, BBC10, BBC16 failed cold rolling before XRD).
 929 Of the 33 FCC-confirmed alloys, most proceeded to full characterization. A few
 930 FCC-confirmed alloys (BBA09, BBB10, BBB16) showed mechanical anomalies
 931 (brittle fracture or processing-induced defects) that limited the quality of their ob-

932 jective data but did not change their phase classification.

933 *Sigma-phase failures* decreased from 0 in BBA to 5 in BBB to 3 in BBC. The
934 reduction from 5 to 3 between BBB and BBC reflects two mid-campaign interven-
935 tions: the transition from TCHEA6 to TCHEA7 with dual-temperature screening,
936 and the retraining of the GPC feasibility classifier on the 5 sigma outcomes from
937 BBB. The GPC and updated CALPHAD screening reduced the sigma failure rate
938 despite the BO continuing to target V-rich, high-strength compositions near the
939 FCC/sigma boundary.

940 *Manufacturing and processing failures* (alloys that never reached XRD verification
941 or that failed during cold rolling) went from 1 in BBA to 0 in BBB to 6 in BBC. The
942 six BBC processing failures all failed cold rolling. The current framework has no
943 mechanism to predict or prevent these failures, as the GPC is trained only on phase
944 outcomes. Post-hoc analysis suggests that V-containing nano-precipitates may be
945 responsible for embrittlement in certain compositions, but this finding postdates
946 the present study and will be reported separately.

947 This observation highlights a deeper tension in this alloy system. Table 5 quantifies
948 the relationship between V content and outcome across all 48 alloys: no alloy with
949 $V = 0$ failed, while failure rates reach 100% at $V = 16$ at.% and stay above 50% for
950 $V \geq 16$ at.%. The top row of the performance landscape ($V \geq 20$ at.%) contains
951 both the three strongest alloys in the campaign (BBC04, BBC02, BBB01—all at V
952 $= 24$ at.%) and four of the eight sigma-phase failures. At $V = 24$ at.%, half the alloys
953 fail and half are the best performers—the definition of a boundary-adjacent opti-
954 mum. High V content simultaneously increases the risk of sigma-phase formation
955 and processing-induced embrittlement. The highest-performing alloys therefore
956 reside adjacent to multiple feasibility boundaries.

957 The failure rate is non-monotonic in V: it climbs to 100% at $V = 16$ at.% then falls
958 back to 80% at $V = 20$ at.% and 50% at $V = 24$ at.%. This is not evidence that
959 V becomes less destabilizing above 16 at.%, it is evidence of surrogate-driven
960 co-selection. By the BBB and BBC iterations, the GP had learned the V–Ni–Cr
961 coupling (Ni stabilizes FCC at high V through the VEC effect; Cr promotes sigma),
962 so the high-V candidates sampled in later iterations were preferentially those with
963 high Ni and low Cr—compositional signatures that survive at the boundary. The
964 $V = 16$ at.% alloys, sampled earlier, did not benefit from this learned filtering. The
965 100% failure rate at $V = 16$ at.% is therefore an artifact of when those samples
966 were queried, not of any special instability at that composition. What the BO is
967 doing at $V = 24$ at.% is hugging the constraint from the feasible side.

Table 5. Failure rate and average yield strength by vanadium content across all 48 alloys.

V (at.%)	Total	Failed	Failure Rate	Notable Alloys
0	8	0	0%	—
4	10	1	10%	BBA04 (manufacturing)
8	6	1	17%	BBC05 (cold rolling)
12	9	2	22%	BBC06, BBC07 (cold rolling)
16	4	4	100%	BBB09, BBB11 (sigma); BBC10, BBC16 (cold rolling)
20	5	4	80%	BBB06, BBB13, BBC08 (sigma); BBC09 (cold rolling)
24	6	3	50%	BBB05, BBC03, BBC14 (sigma); BBC04, BBC02, BBB01 (FCC, top 3 UTS)

968 To quantify whether the relationship between V content and infeasibility is sta-
 969 tistically significant, we applied two complementary tests [49]. First, treating V
 970 content as a continuous variable, the Pearson correlation between V content and
 971 infeasibility (coded as 0 or 1) is $r = 0.54$ ($P = 9.0 \times 10^{-5}$, $n = 48$): higher V is as-
 972 sociated with higher probability of failure, and the probability that this association
 973 arises by chance is less than 0.01%. This association is driven by V specifically,
 974 not by the V + Cr combination that defines the sigma stability threshold: V and
 975 Cr are uncorrelated across the 48 alloys ($r = 0.06$), confirming that V is a dom-
 976 inant predictor of infeasibility. Second, dividing alloys into high-V (≥ 16 at.%)
 977 and low-V (< 16 at.%) groups, 11 of 15 high-V alloys were infeasible compared to
 978 4 of 33 low-V alloys—an odds ratio of 19.9 (Fisher’s exact test, $P = 5.3 \times 10^{-5}$),
 979 meaning a high-V alloy is approximately 20 times more likely to fail than a low-V
 980 alloy. This association is robust to the choice of threshold: the odds ratio remains
 981 significant ($P < 0.05$) for all cutoffs from $V \geq 12$ to $V \geq 20$ at.%.

982 Among the 30 feasible alloys with yield strength data, V content correlates with
 983 YS at $r = 0.84$ ($P = 8.7 \times 10^{-9}$). The same element that most strongly predicts
 984 mechanical performance also most strongly predicts failure—both at significance
 985 levels below $P < 10^{-4}$. The BO’s preferential sampling of high-V alloys in itera-
 986 tions BBB and BBC (15 of 32 alloys with $V \geq 16$ at.%, compared to 0 of 16 in the
 987 diversity-sampled BBA) is itself evidence that the surrogates correctly identified V
 988 as the performance driver. The selection bias toward high V is not a confound—it
 989 is the optimizer doing its job, concentrating budget where the predicted payoff is
 990 highest. The consequence is that both the best alloys and the most failures accu-

991 mulate in the same compositional region, which is the signature of a boundary-
992 adjacent optimum.

993 This dual role is a pattern consistent with the general observation in constrained
994 optimization that optimal solutions tend to lie on or near active constraint bound-
995 aries [50]. This coupling arises when the same compositional variable drives both
996 performance and constraint violation: increasing V improves strength through lat-
997 tice distortion effects (the objective gradient points toward high V) but simultane-
998 ously promotes sigma-phase formation (the constraint gradient points in the same
999 direction). The unconstrained optimum would be a very high-V alloy that forms
1000 sigma. The constrained optimum is therefore the alloy with the highest V content
1001 that remains single-phase FCC—on the phase stability boundary by definition.

1002 (Formally, the Karush–Kuhn–Tucker conditions for constrained optimality require
1003 $\nabla f + \lambda \nabla g = 0$ with multiplier $\lambda > 0$ at the solution, placing the optimum on
1004 the active constraint boundary $g(\mathbf{x}) = 0$ [36]. The KKT conditions apply strictly
1005 to continuous, single-objective problems; the present campaign operates on a dis-
1006 crete compositional grid with multiple objectives, but the underlying intuition—
1007 that aligned objective and constraint gradients push optima toward boundaries—
1008 transfers to this setting.) If the optimum were far from the constraint, the constraint
1009 would not be restricting the search, and the problem would be effectively uncon-
1010 strained. This has a practical implication for campaign design: in alloy systems
1011 where the primary strengthening element also destabilizes the target phase (e.g., V
1012 in the present system), a feasibility-aware acquisition strategy is a prerequisite for
1013 accessing the high-performing region of the design space. The present campaign
1014 provides an experimental realization of this principle: vanadium drives both the
1015 objective ($r = 0.84$ with YS) and the constraint ($r = 0.54$ with infeasibility), and
1016 the three strongest alloys in the campaign share the V = 24 at.% boundary with
1017 three sigma failures. The boundary phenomenon has been observed implicitly in
1018 metastability-engineered HEAs, where optimal TRIP-mediated strength–ductility
1019 combinations arise at the edge of FCC phase stability [9], and has been proposed
1020 as an algorithmic strategy for constrained Bayesian optimization [37, 38]. How-
1021 ever, the explicit connection between performance–constraint coupling through a
1022 shared compositional variable and the KKT prediction that constrained optima lie
1023 on active boundaries has not, to our knowledge, been demonstrated quantitatively
1024 in an experimental alloy campaign.

1025 The sigma-phase trend ($0 \rightarrow 5 \rightarrow 3$) is the more informative metric for evaluating
1026 the feasibility-aware acquisition strategy. In iteration BBA, candidates were se-

1027 lected by diversity-aware k-medoids sampling across the interior of the FCC stabil-
1028 ity region, producing no sigma failures. In BBB, the BO targeted high-V, high-Cr
1029 regions where the GP surrogates predicted strong mechanical performance—the
1030 same regions where sigma is thermodynamically competitive. The 5 sigma failures
1031 provided the GPC with its first substantial training data and triggered the TCHEA7
1032 transition. In BBC, the updated screening and retrained GPC reduced sigma fail-
1033 ures to 3 despite continued exploration of the V-rich boundary—evidence that the
1034 feasibility-aware acquisition learned to avoid the most sigma-prone compositions
1035 while preserving access to the high-performing V-rich region.

1036 The counterfactual remains instructive. A hard-filtered strategy would have ex-
1037 cluded the high-V region entirely after the BBB sigma failures. BBC04 and BBC02—
1038 the two highest-UTS alloys in the campaign—would never have been synthesized.
1039 The BRAVE framework found them by accepting controlled risk near the phase
1040 boundary, with the GPC progressively sharpening the feasibility boundary between
1041 iterations.

1042 The relevant performance metric is the performance gained per iteration, not the
1043 absolute failure rate. Despite 9/16 total failures in BBC (3 sigma + 6 process-
1044 ing), the 7 feasible alloys from that iteration included the strongest compositions
1045 in the campaign (BBC04, BBC02), and the hypervolume continued to increase
1046 (4.7% gain over BBB). Extending the feasibility model to incorporate processing
1047 constraints—predicting which compositions are likely to fail cold rolling, inde-
1048 pendent of phase stability—is a priority for future campaigns.

1049 The feasibility analysis also exposes CALPHAD database fidelity as a practical
1050 bottleneck for closed-loop alloy campaigns. The TCHEA6 database, used for ini-
1051 tial screening, failed to predict sigma formation in several V- and Cr-rich com-
1052 positions that were experimentally confirmed as multiphase. The mid-campaign
1053 transition to TCHEA7—with its expanded sigma, BCC, HCP, and liquid phase as-
1054 sessments across >100 ternary subsystems—reduced but did not eliminate these
1055 prediction errors (Section 3.1). The GPC feasibility classifier absorbs residual
1056 prediction errors by learning the empirical feasibility boundary from experimental
1057 phase outcomes, but its effectiveness depends on the quality of the thermodynamic
1058 screening that defines the initial candidate pool. A poorly calibrated CALPHAD
1059 database shifts the entire feasible space, potentially excluding high-performing re-
1060 gions from the outset or flooding the candidate pool with infeasible compositions
1061 that consume experimental budget before the GPC has sufficient training data to
1062 compensate. This dependency has practical implications: future campaigns should

(i) validate CALPHAD predictions against a small exploratory batch before committing to full-scale optimization, (ii) incorporate dual-temperature screening (as we adopted here after iteration BBB) from the first iteration, and (iii) prioritize thermodynamic databases with assessed coverage of the relevant ternary and quaternary subsystems.

4. Conclusions

In this work, we applied a risk-aware Bayesian optimization campaign (BRAVE) to the Al–V–Cr–Mn–Fe–Co–Ni–Cu FCC HEA system, synthesizing 48 alloys over three closed-loop iterations. We optimized five objectives simultaneously (YS, UTS/YS, strain at UTS, $H_{\text{dyn}}/H_{\text{qs}}$, DoP) while exploring 0.12% of the 27,240-composition feasible space.

1. **The highest-performing compositions reside on the FCC phase stability boundary.** Vanadium drives both yield strength ($r = 0.84$, $n = 30$ feasible alloys) and sigma-phase formation ($r = 0.54$, $P = 9 \times 10^{-5}$, $n = 48$). At V = 24 at.%, the three strongest alloys and three sigma failures share the same compositional point—the definition of a boundary-adjacent optimum. This outcome corresponds to the KKT prediction that constrained optima lie on active boundaries when objective and constraint gradients align [36, 50]. Although such optima have been observed implicitly in metastability-engineered HEAs [9] and proposed algorithmically [37, 38], they have not, to our knowledge, been demonstrated with statistical significance in a closed-loop experimental campaign.
2. **BRAVE navigated the boundary where simpler strategies cannot.** Hard filtering would have excluded the V-rich region after the first sigma failures. BBC04 and BBC02 (the two highest-UTS alloys) would never have been synthesized. Conversely, feasibility-blind BO would have repeatedly queried infeasible compositions without learning to avoid them. BRAVE embedded a learned feasibility probability into EHVI, concentrating resources in the V-rich, Ni-rich corner (<0.4% of the feasible space, $P \approx 6.5 \times 10^{-6}$ under random sampling) while reducing the sigma rate from 5/16 to 3/16. Processing failures ($1 \rightarrow 0 \rightarrow 6$) lie outside the framework’s scope. Despite 9/16 total infeasible alloys in BBC, the 7 feasible compositions included the strongest in the campaign. A computational benchmark (Appendix) confirms that risk-weighted EHVI matches oracle performance with $\sim 5\times$ fewer wasted evaluations.

- 1098 **3. Two compositional regimes emerged from the 8-element space.** A high-
1099 strength regime ($V = 20\text{--}24$ at.%, $\text{Cr} \leq 4$ at.%, $\text{Ni} = 36\text{--}60$ at.%) produced
1100 BBC04 (~ 1480 MPa UTS, 50% elongation), where Ni stabilizes FCC at ex-
1101 treme V concentrations through VEC effects. A high-ductility regime ($V =$
1102 $8\text{--}12$ at.%, $\text{Mn} = 16$ at.%) achieved UTS/YS up to 4.20 at $>50\%$ elongation,
1103 consistent with SFE reduction and enhanced strain hardening [9, 10]. Nei-
1104 ther regime was accessible in Campaign 1’s 6-element system. SHAP anal-
1105 ysis confirmed V as the most influential feature (highest mean |SHAP|), con-
1106 sistent with its strong linear correlation with strength ($r = 0.82$). Ni, ranked
1107 fourth by mean |SHAP|, shows near-zero linear correlation with strength ($r =$
1108 -0.03) but non-negligible SHAP contributions whose sign reverses with
1109 compositional context—consistent with interaction-mediated, non-monotonic
1110 influence. Cu correlated negatively with all strength metrics and positively
1111 with DoP.
- 1112 **4. The campaign avoided the strength–ductility trade-off and compounded**
1113 **gains from Campaign 1.** Median YS increased across iterations while duc-
1114 tility was preserved, and inter-quartile ranges narrowed—indicating conver-
1115 gence. Every single-phase FCC alloy exceeded Campaign 1’s best UTS/YS
1116 (2.93 [46]): minimum 3.07, median 3.79, 9 above 4.0. We attribute this
1117 to three compounding factors: (i) informative priors from Campaign 1 data
1118 and the Varvenne–Luque–Curtin model, (ii) a design space expanded based
1119 on Campaign 1’s lessons, and (iii) objectives refined from the finding that
1120 hardness alone is an incomplete proxy for mechanical response.
- 1121 **5. DoP operated as a simulation-in-the-loop objective.** Calibrated from ten-
1122 sile and SHPB data via a Cowper–Symonds model, DoP tracks YS ($r =$
1123 -0.94) but also captures rate-sensitivity effects invisible to quasi-static test-
1124 ing. The strongest alloys (high V) showed lower rate-dependent strengthen-
1125 ing, consistent with dislocation-glide-dominated deformation—a trade-off
1126 that is invisible without a rate-sensitive objective.
- 1127 **6. CALPHAD database fidelity limits the feasibility classifier.** The mid-
1128 campaign TCHEA6 $\rightarrow$ TCHEA7 transition reduced but did not eliminate
1129 false negatives—the errors that produce failed experiments. The GPC learns
1130 the empirical boundary from phase outcomes. A poorly calibrated database
1131 can still exclude high-performing regions or admit too many infeasible com-
1132 positions before the GPC acquires sufficient training data.

1133 We note several limitations. The campaign optimized composition only; microstruc-
1134 ture and processing were not design variables. Variability in grain size ($10\text{--}35\ \mu\text{m}$)

1135 and recrystallization response introduced scatter that compositional surrogates can-
1136 not capture—several phase-pure alloys showed anomalous tensile behavior trace-
1137 able to incomplete recrystallization, reducing the signal-to-noise available to the
1138 GP models. The DoP model uses a Cowper–Symonds framework without damage
1139 criteria, limiting its scope to relative penetration ranking. Furthermore, the GPC
1140 (trained on 8 sigma failures) cannot predict the 7 processing failures (i.e., cold-
1141 rolling defects) that dominated BBC’s budget—a distinct failure mode outside the
1142 classifier’s design scope.

1143 The three-iteration budget was set by experimental throughput rather than conver-
1144 gence criteria, and the diminishing HV gain (24.2% in BBB, 4.7% in BBC) leaves
1145 open whether additional iterations would yield substantial further improvement.

1146 In future work, we plan to incorporate microstructural descriptors (e.g., grain size,
1147 texture, precipitate fraction) as design variables and extend the feasibility model to
1148 processing failures. Testing BRAVE on systems with more complex phase com-
1149 petition (e.g., refractory HEAs with BCC/HCP/sigma boundaries) would assess
1150 generalizability. The KKT observation predicts boundary-adjacent optima should
1151 recur whenever the performance-driving element also destabilizes the target phase.

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1183 **Declaration of competing interest**

1184 The authors declare that they have no known competing financial interests or per-
1185 sonal relationships that could have appeared to influence the work reported in this
1186 paper.

1187 **Data Availability**

1188 The data generated and analyzed during this study is included in Supplementary Information
1189 and is also available in the github repository below, provided as an Excel database.

1190 **Code Availability**

1191 Code used in this project is available on <https://github.com/mrinalinimulukutla/BRAVE>.

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1210 **Appendix**

1211 *A. Mathematical Details of the BRAVE Framework*

1212 This section collects the mathematical details of the surrogate modeling, multi-
 1213 objective acquisition, multi-source fusion, and batch selection components sum-
 1214 marized in Section 2.6.

1215 *A.1 Gaussian Process Regression*

1216 In Bayesian optimization, probabilistic surrogate models are used to represent
 1217 expensive-to-evaluate objective functions and guide candidate selection through
 1218 uncertainty-aware predictions. We used Gaussian process regression (GPR) as the
 1219 surrogate due to its low computational cost and flexible distance-based correlation
 1220 structure [19].

1221 Assuming there are N previously observed data denoted by $\{\mathbf{X}_N, \mathbf{y}_N\}$, where $\mathbf{X}_N =$
 1222 $(\mathbf{x}_1, \dots, \mathbf{x}_N)$ and $\mathbf{y}_N = (f(\mathbf{x}_1), \dots, f(\mathbf{x}_N))$, then the GP probabilistic prediction at
 1223 unobserved location $\mathbf{x}$ is a normal distribution:

$$f_{\text{GP}}(\mathbf{x}) \mid \mathbf{X}_N, \mathbf{y}_N \sim \mathcal{N}(\mu(\mathbf{x}), \sigma_{\text{GP}}^2(\mathbf{x})) \quad (4)$$

1224 where

$$\begin{aligned} \mu(\mathbf{x}) &= K(\mathbf{X}_N, \mathbf{x})^T [K(\mathbf{X}_N, \mathbf{X}_N) + \sigma_n^2 I]^{-1} \mathbf{y}_N \\ \sigma_{\text{GP}}^2(\mathbf{x}) &= k(\mathbf{x}, \mathbf{x}) - K(\mathbf{X}_N, \mathbf{x})^T [K(\mathbf{X}_N, \mathbf{X}_N) + \sigma_n^2 I]^{-1} K(\mathbf{X}_N, \mathbf{x}) \end{aligned} \quad (5)$$

1225 here, k is a real-valued kernel function to calculate correlations, $K(\mathbf{X}_N, \mathbf{X}_N)$ as a
 1226 $N \times N$ matrix with m, n entry as $k(\mathbf{x}_m, \mathbf{x}_n)$, and $K(\mathbf{X}_N, \mathbf{x})$ is a $N \times 1$ vector with m^{th}
 1227 entry as $k(\mathbf{x}_m, \mathbf{x})$. The term σ_n^2 is used to model experiment errors. A common

choice for kernel function is squared exponential:

$$k(\mathbf{x}, \mathbf{x}') = \sigma_s^2 \exp \left(- \sum_{h=1}^d \frac{(x_h - x'_h)^2}{2l_h^2} \right) \quad (6)$$

where d is the dimensionality of the input space, σ_s^2 is the signal variance, and l_h , where $h = 1, 2, \dots, d$, is the characteristic length-scale to determine correlation strength between data points within dimension h .

A.2 Informative Priors

Although standard GPR sets the mean function to zero, this forfeits the opportunity to encode prior knowledge. An informative prior instead sets the mean function to a physics-based model prediction, $\mu(\mathbf{x})$, and trains the GPR on the discrepancy $f'(\mathbf{x}) = f_{\text{exp}}(\mathbf{x}) - \mu(\mathbf{x})$ between experimental observations and the prior. The posterior $f = f' + \mu$ defaults to the physics-based prediction where no experimental data exist, and overrides it where observations are available.

In this work, we used GPR with an informative prior that integrates physical insights from the Varvenne–Luque–Curtin solid solution strengthening model [48] as the prior, μ . This model captures the strengthening behavior of FCC HEAs by accounting for solute misfit volumes, modulus mismatch, and local lattice distortions in a fully random solution. We combined this physics-based prior with limited experimental data, including Vickers hardness (HV) and temperature-dependent yield strength (YS) measurements.

Let the true composition-temperature-dependent mechanical behavior of the HEAs be denoted by $f_{\text{exp}}(\mathbf{x})$, where $\mathbf{x}$ is a vector describing the alloy composition and test temperature. The GPR model is built over the residual between experimental observations and the prior prediction, $f_{\text{dis}}(\mathbf{x}) = f_{\text{exp}}(\mathbf{x}) - \mu(\mathbf{x})$. The final GPR prediction is then:

$$f(\mathbf{x}) = f_{\text{dis}}(\mathbf{x}) + \mu(\mathbf{x}).$$

The machine learning models were developed using data from [46] in conjunction with the CBFV [51] and HEACalculator packages [52] for composition feature generation. These tools generated over 5000 features for each composition, which were subsequently optimized using the RecursiveByShapValues method [53] to eliminate redundant features based on SHAP values. Given the tabular nature of the dataset, with a limited number of datapoints and a high dimensionality of features, the CatBoost gradient boosting algorithm [53] was used.

1258 A.3 Multi-Objective Optimization via EHVI

1259 Because EHVI is implemented in minimization form, the four desirable-to-maximize
 1260 objectives (YS, UTS/YS, strain at UTS, and $H_{\text{dyn}}/H_{\text{qs}}$) were sign-inverted before
 1261 surrogate modeling and hypervolume calculation, while DoP remained in its orig-
 1262 inal minimization form.

1263 A multi-objective optimization problem is defined as

$$\text{minimize } \{f_1(\mathbf{x}), \dots, f_n(\mathbf{x})\}, \mathbf{x} \in \mathcal{X} \quad (7)$$

1264 where $f_1(\mathbf{x}), \dots, f_n(\mathbf{x})$ are the objectives and $\mathcal{X}$ is the feasible design space. Typ-
 1265 ically, no single solution simultaneously optimizes all objectives. The solution is
 1266 instead a set of non-dominated designs with trade-offs, forming the Pareto front in
 1267 the objective space. The optimal solutions $\mathbf{y}$ are denoted as $\mathbf{y} < \mathbf{y}'$ and expressed
 1268 as

$$\begin{aligned} \{\mathbf{y} : \mathbf{y} = (y_1, y_2, \dots, y_n), y_i \leq y'_i \ \forall i \in \{1, 2, \dots, n\}, \\ \exists j \in \{1, 2, \dots, n\} : y_j < y'_j\} \end{aligned} \quad (8)$$

1269 where $\mathbf{y}' = (y'_1, y'_2, \dots, y'_n)$ denotes any possible objective output.

1270 Several approaches exist for approximating the Pareto front, including weighted
 1271 sum [54], adaptive weighted sum [55], normal boundary intersection [56], and hy-
 1272 pervolume indicator methods [57–63]. Hypervolume is the space bounded by a
 1273 fixed reference point and the current Pareto front approximation. Maximizing it
 1274 pushes the approximation toward the true Pareto front. The acquisition function
 1275 used here is Expected Hypervolume Improvement (EHVI), computed via the re-
 1276 cursive decomposition method of Zhao et al. [22]. Following [64], EHVI is defined
 1277 as

$$\mathbb{E}[\text{HVI}(\mathbf{y})] = \int_U \mathbb{P}(\mathbf{y} < \mathbf{y}') d\mathbf{y}' \quad (9)$$

1278 where $\mathbb{P}(\mathbf{y} < \mathbf{y}')$ is the probability that $\mathbf{y}'$ dominates $\mathbf{y}$ and U is the dominated
 1279 hypervolume. Given an independent GP for each objective, the posterior predictive
 1280 is $y_i \sim \mathcal{N}(\mu_i, \sigma_i^2)$ where $i \leq m$ and μ_i, σ_i^2 are the mean and variance of the i^{th}
 1281 objective. For a new candidate:

$$\mathbb{P}(\mathbf{y} < \mathbf{y}') = \prod_{i=1}^m \Phi\left(\frac{y'_i - \mu_i}{\sigma_i}\right) \quad (10)$$

1282 where Φ is the CDF of the standard normal. Details regarding the closed-form

1283 expression of Eq. (9) can be found in Refs. [22, 64–67].

1284 A.4 Multi-Source Modeling via Reification

1285 Four objectives (YS, UTS/YS, strain at UTS, $H_{\text{dyn}}/H_{\text{qs}}$) were measured experimen-
 1286 tally. The fifth (DoP) was a simulation output derived from FEM models calibrated
 1287 to the tensile and SHPB data acquired in the same iteration. DoP therefore passed
 1288 through a two-level model chain—experiment $\rightarrow$ constitutive calibration $\rightarrow$ FEM
 1289 $\rightarrow$ GP—and its uncertainty compounded surrogate approximation error with up-
 1290 stream calibration error. All five objectives entered the EHVI computation as GP
 1291 posterior distributions regardless of their origin.

1292 To avoid assumptions about the hierarchy of information sources, we estimated
 1293 inter-source correlations using the reification technique [68, 69]. Following [70],
 1294 the fused mean and variance are:

$$\mathbb{E}[\hat{f}(\mathbf{x})] = \frac{\mathbf{e}^T \tilde{\Sigma}(\mathbf{x})^{-1} \boldsymbol{\mu}(\mathbf{x})}{\mathbf{e}^T \tilde{\Sigma}(\mathbf{x})^{-1} \mathbf{e}} \quad (11)$$

1295

$$\text{Var}(\hat{f}(\mathbf{x})) = \frac{1}{\mathbf{e}^T \tilde{\Sigma}(\mathbf{x})^{-1} \mathbf{e}} \quad (12)$$

1296 where $\mathbf{e} = [1, \dots, 1]^T$, $\boldsymbol{\mu}(\mathbf{x}) = [\mu_1(\mathbf{x}), \dots, \mu_m(\mathbf{x})]^T$ given m information sources, and
 1297 $\tilde{\Sigma}(\mathbf{x})$ is the covariance matrix with diagonal elements σ_i^2 and off-diagonal elements
 1298 $\bar{\rho}_{ij}\sigma_i\sigma_j$. The correlation coefficient $\bar{\rho}_{ij}$ is calculated via reification:

$$\bar{\rho}_{ij}(\mathbf{x}) = \frac{\sigma_j^2(\mathbf{x})}{\sigma_i^2(\mathbf{x}) + \sigma_j^2(\mathbf{x})} \rho_{ij}(\mathbf{x}) + \frac{\sigma_i^2(\mathbf{x})}{\sigma_i^2(\mathbf{x}) + \sigma_j^2(\mathbf{x})} \rho_{ji}(\mathbf{x}) \quad (13)$$

1299 where

$$\begin{aligned} \rho_{ij}(\mathbf{x}) &= \frac{\sigma_i(\mathbf{x})}{\sqrt{(\mu_i(\mathbf{x}) - \mu_j(\mathbf{x}))^2 + \sigma_i^2(\mathbf{x})}}, \\ \rho_{ji}(\mathbf{x}) &= \frac{\sigma_j(\mathbf{x})}{\sqrt{(\mu_i(\mathbf{x}) - \mu_j(\mathbf{x}))^2 + \sigma_j^2(\mathbf{x})}}. \end{aligned} \quad (14)$$

1300 A.5 Batch Bayesian Optimization

1301 GP surrogates depend on kernel length-scale hyperparameters that control corre-
 1302 lation strength across the input space. Although standard tuning methods (e.g.,
 1303 maximum likelihood [19], cross-validation) are effective with sufficient data, they

1304 can become trapped in local optima when training data are sparse [71, 72].

1305 To address this, we constructed a large ensemble of GPs with randomly sampled
 1306 length-scales, each encoding a different smoothness assumption [71]. Each GP
 1307 made a different inference about the underlying objective function, yielding differ-
 1308 ent candidate suggestions:

$$\mathbf{x}_{1:n} = \arg \max_{\mathbf{x} \in \mathcal{Y}} \text{EHVI}^N(\mathbf{x} | \text{GP}(\mathbb{D}_N, \theta_{1:n})) \quad (15)$$

1309 where n Gaussian processes were built with different hyperparameter sets θ given
 1310 data $\mathbb{D}$. From the n candidate suggestions, k-medoids clustering selected a batch of
 1311 compositions that maximized diversity while preserving aggregate hypervolume
 1312 gain.

1313 *B. Benchmarking Risk-Aware Multi-Objective Bayesian Optimization*

1314 To quantitatively assess the effectiveness of the proposed risk-aware multi-objective
 1315 Bayesian optimization (MOBBO) strategy, we constructed a controlled computa-
 1316 tional benchmark based on the six-component Al–V–Cr–Fe–Co–Ni HEA space.
 1317 The full design space was discretized at 4 at.% resolution, generating a pool of
 1318 23,694 distinct compositions with precomputed objective properties and phase sta-
 1319 bility metrics.

1320 Two bi-objective optimization problems were defined: (i) maximize yield strength
 1321 proxy (CV_YS) and Pugh ratio (B/G), and (ii) maximize CV_YS and Cauchy pres-
 1322 sure ($C_{12} - C_{44}$). CV_YS was computed using the Curtin–Varvenne solid solution
 1323 strengthening model [48]. Pugh ratio and Cauchy pressure were approximated via
 1324 rule-of-mixtures aggregation of elemental elastic constants.

1325 Feasibility was introduced through a dual-database configuration that mimics real-
 1326 istic model discrepancy: screening used TCHEA6 ($\text{FCC} \geq 0.99$ at 700°C), while
 1327 ground-truth feasibility labels were assigned from TCHEA7. Any composition
 1328 with TCHEA7 FCC fraction < 0.99 was treated as infeasible.

1329 Four acquisition strategies were compared:

- 1330 1. **Baseline MOBBO**: EHVI applied across the full design space without fea-
 1331 sibility awareness.
- 1332 2. **Hard-Filter MOBBO**: EHVI restricted to TCHEA6-predicted FCC-stable
 1333 compositions.
- 1334 3. **Risk-Weighted MOBBO**: EHVI multiplied by $p_{\text{feas}} = (\text{FCC}_{\text{TCHEA7}})^\alpha$ with
 1335 $\alpha = 2$.

1336 4. **Oracle MOBBO**: EHVI weighted by the true binary feasibility label from
1337 TCHEA7 (upper bound).

1338 Each campaign selected 10 initial compositions via k -medoids clustering, then ran
1339 20 sequential iterations of batch size 10, with GP regressors for objectives and a
1340 Random Forest classifier for feasibility. Five random seeds yielded 100 total cam-
1341 paigns per objective pair. Performance was evaluated using feasible hypervolume
1342 (HV), Pareto front size, and feasibility violation rate (FVR).

1343 [Figure 20](#) summarizes the results. The differences between strategies are consistent
1344 across both objective pairs:

- 1345 • **Baseline MOBBO** fails completely: near-zero feasible HV, trivial Pareto
1346 sets, $FVR \approx 1.0$, with $>90\%$ of evaluations wasted on infeasible candidates.
- 1347 • **Hard-Filter MOBBO** improves feasibility ($FVR \approx 0.15$) but plateaus early
1348 when TCHEA6 mispredictions over-restrict exploration. It achieves HV
1349 = 7.25 (Pugh) and 4.66 (Cauchy) but lacks robustness near misclassified
1350 boundaries.
- 1351 • **Risk-Weighted MOBBO** matches the oracle in final HV (7.25 and 4.66),
1352 maintains low FVR (≈ 0.15 – 0.20), and converges within 3–5 iterations. By
1353 penalizing rather than excluding uncertain candidates, it preserves explo-
1354 ration flexibility near critical boundaries.
- 1355 • **Oracle MOBBO** achieves identical final HV but wastes ≈ 170 – 190 evalua-
1356 tions on infeasible candidates early on, demonstrating that feasibility knowl-
1357 edge alone is insufficient—it must be incorporated into the acquisition func-
1358 tion.

1359 The three feasibility-aware strategies converge to the same final HV because that
1360 value is the ceiling: it is the true feasible Pareto front of the discretized 6-component
1361 space, and any strategy that avoids wasting the budget on infeasibles will even-
1362 tually saturate at it. Final HV therefore cannot discriminate among Hard-Filter,
1363 Risk-Weighted, and Oracle. What separates them is the path to that ceiling. Hard-
1364 Filter reaches it in this benchmark because the accessible optimum happens to sit
1365 safely inside TCHEA6’s predicted feasible set; in a system where the optimum lies
1366 on a TCHEA6-misclassified boundary—which is the situation of the 8-element
1367 experimental campaign in the main text—Hard-Filter would exclude the winning
1368 region outright. Risk-Weighted reaches the same ceiling without depending on

a ground-truth feasibility database: it learns the empirical boundary from experimental outcomes as they arrive, matching Oracle’s final HV with $\sim 5\times$ fewer wasted evaluations. The binding bottleneck in feasibility-constrained MOBBO is therefore feasibility awareness. Baseline MOBBO’s failure demonstrates this: EHVI is a competent acquisition function, but without feasibility awareness it wastes the budget on infeasible candidates and the GPs never accumulate enough clean signal to converge. Among feasibility-aware strategies, the operationally important distinction is robustness to database discrepancy and resource efficiency, not peak Pareto quality.

Even with perfect feasibility knowledge, the oracle still queries ≈ 170 – 190 infeasible high-objective candidates over the 20-iteration budget, because feasibility labels alone do not suppress the acquisition function’s attraction to unexplored regions with high predicted performance. Risk-weighted MOBBO avoids this by embedding feasibility directly into the acquisition score, matching the oracle’s final HV with $\sim 5\times$ fewer wasted evaluations. The experimental campaign shows the same pattern: identifying the feasibility boundary is necessary but not sufficient—the acquisition function must be designed to navigate it.

The magnitude of this efficiency gain is not compatible with chance. Under a Poisson null model in which risk-weighted MOBBO and the oracle have the same per-iteration probability of wasting an evaluation, the probability of observing ≈ 35 or fewer wasted evaluations from risk-weighted MOBBO given the oracle’s baseline of ≈ 180 wasted evaluations is $P < 10^{-30}$ per bi-objective task (normal approximation to the Poisson tail, $z \approx -10.8$), so $P < 10^{-60}$ jointly across the CV(YS)–Pugh and CV(YS)–Cauchy pressure tasks. In parallel with the hypergeometric bound reported in Section 3.4 for the experimental campaign ($P \approx 6.5 \times 10^{-6}$ that the four highest-YS alloys cluster in the V-rich, Ni-rich corner by chance), the ablation therefore establishes the risk-aware advantage on two complementary axes: statistical improbability of the observed concentration in the experimental campaign, and statistical improbability of the observed efficiency gain in the computational benchmark.

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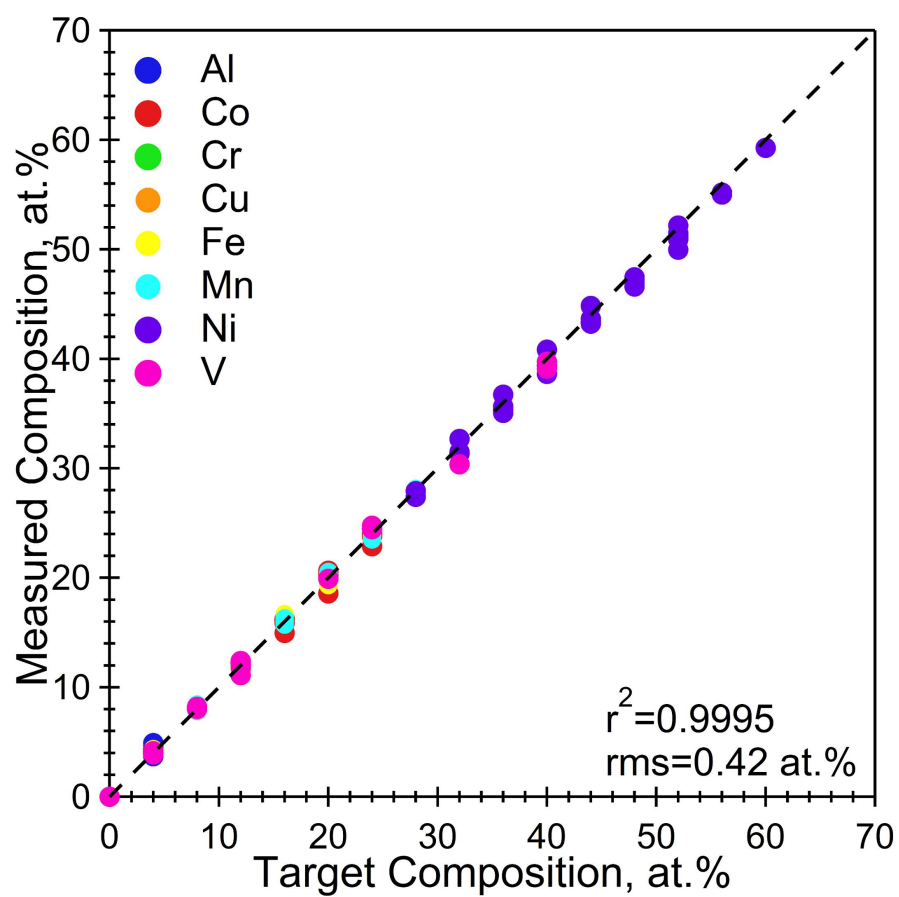


Figure 5. Comparison between target and realized chemistry for the samples generated in this study.

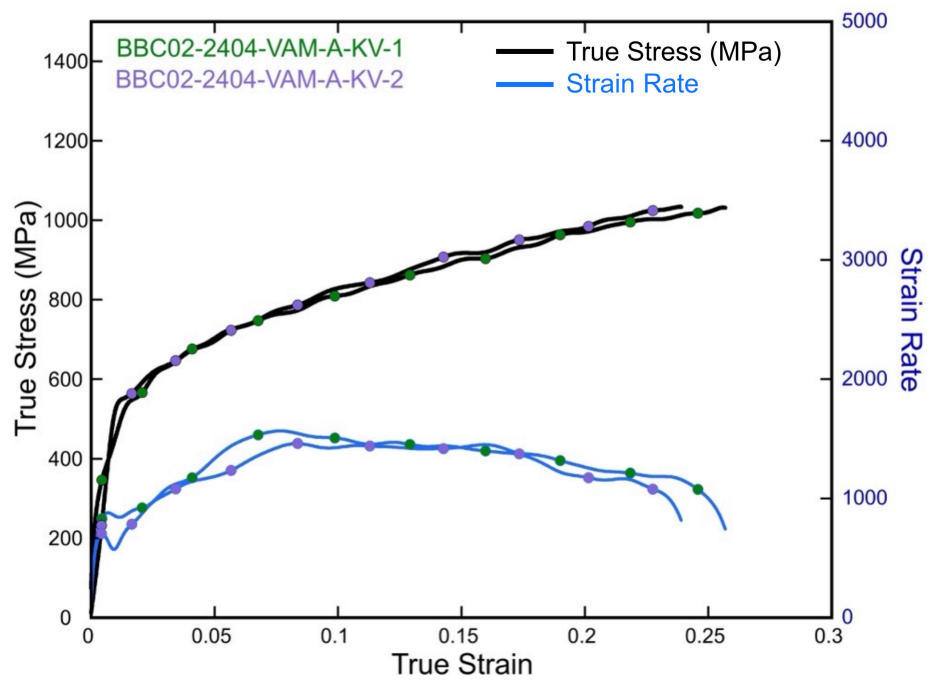


Figure 6. Dynamic true stress–strain responses (black) and corresponding strain rate as a function of strain (blue) for two HEA samples tested at 1500 s^{-1} .

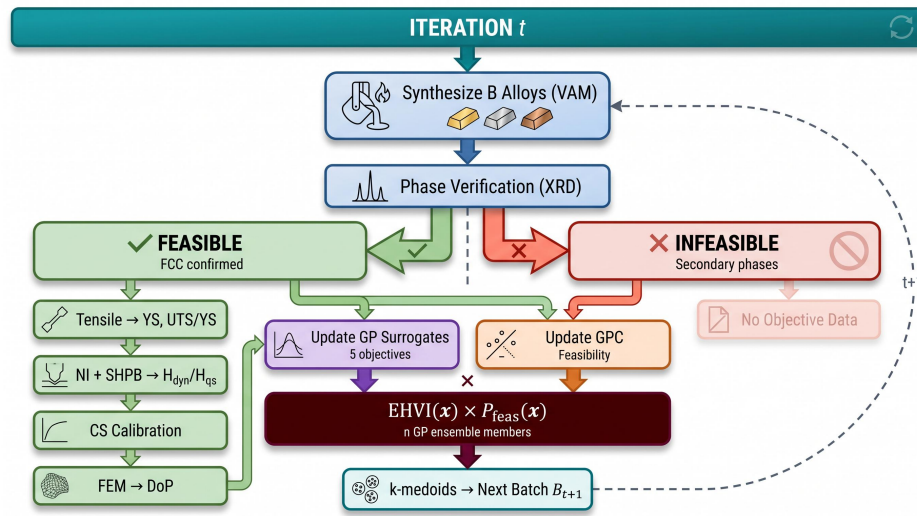


Figure 7. Data flow architecture for one iteration of the BRAVE campaign. Phase verification (XRD) classifies each synthesized alloy as feasible or infeasible, creating two parallel update paths: feasible alloys provide objective measurements for GP surrogate training, while all phase outcomes (feasible and infeasible) update the GPC feasibility classifier. The paths converge at the acquisition step, where EHVI is scaled by the feasibility probability $P_{\text{feas}}(\mathbf{x})$ to penalize high-risk candidates. k-medoids clustering selects the next batch from the top-ranked candidates.

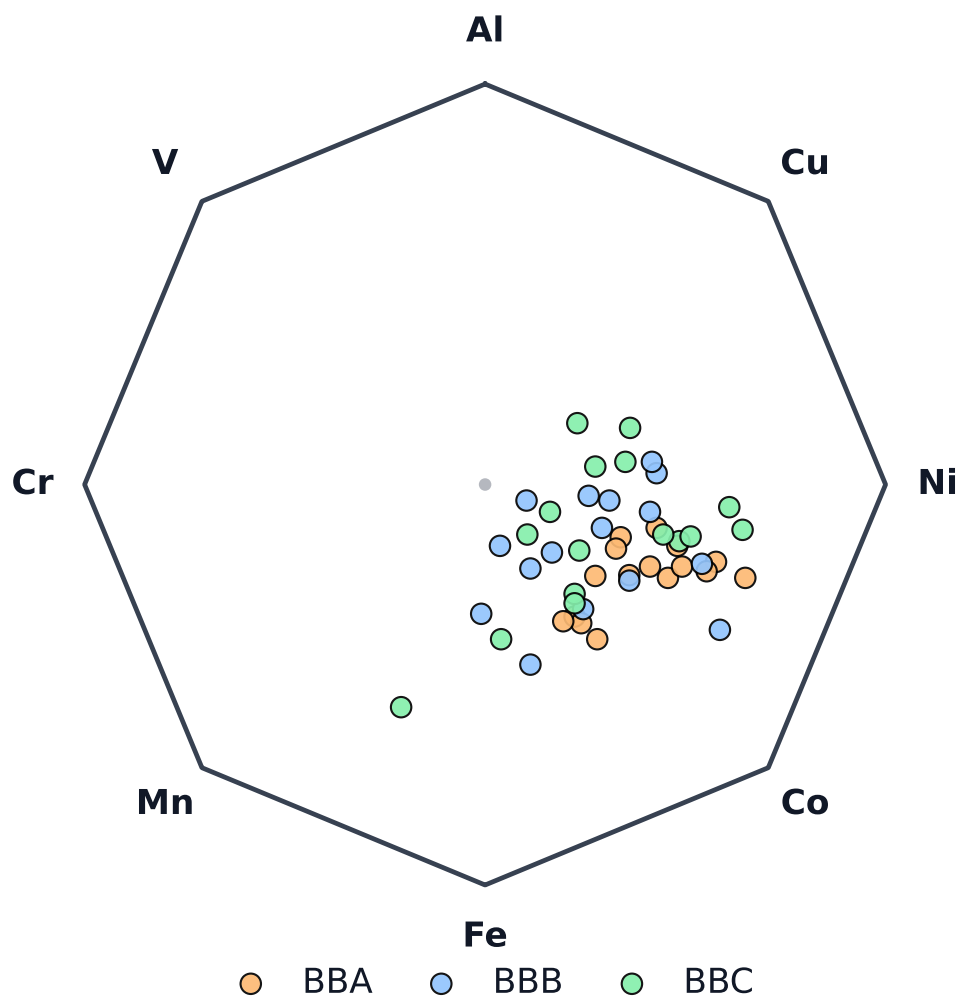


Figure 8. Affine projection of the target chemistries of all 48 alloys synthesized in this study onto a regular octagon representing the 8-element composition space (Al–V–Cr–Mn–Fe–Co–Ni–Cu). Each point is the affine-coordinate projection of one alloy composition and is colored by design iteration.

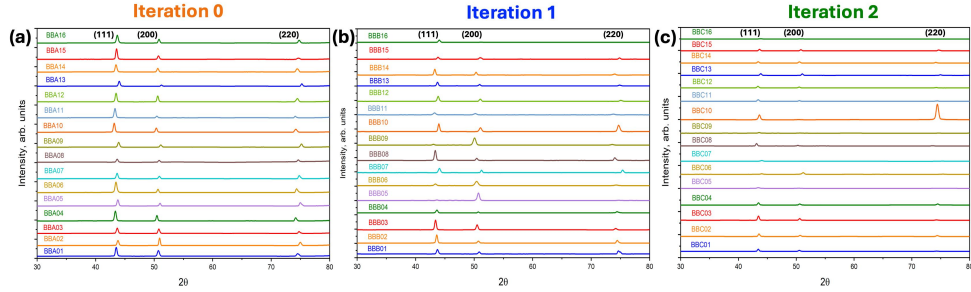


Figure 9. X-ray diffraction (XRD) patterns for all alloys across 3 iterations (a–c), showing FCC spectral peaks at (111), (200), and (220) planar reflections. Alloys with secondary sigma phase are indicated.

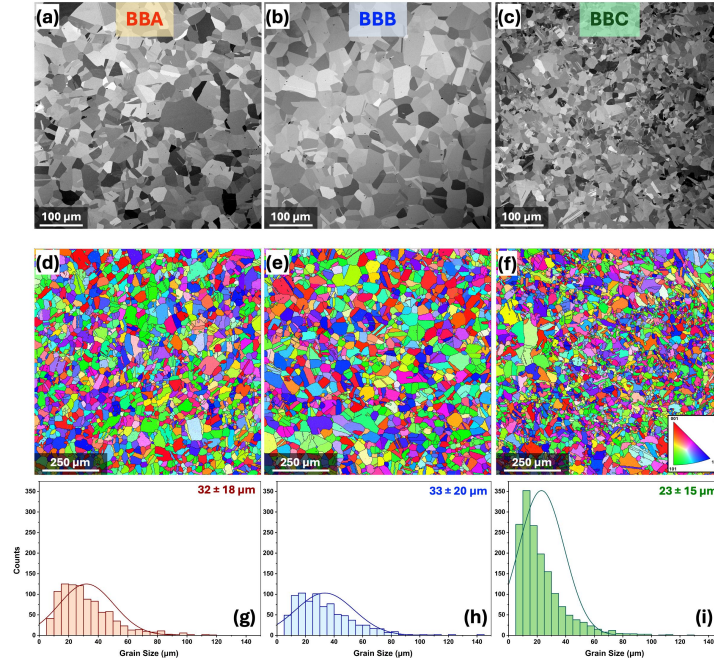


Figure 10. Microstructure of representative FCC alloy compositions across 3 iterations. (a–c) SEM-BSE micrographs showing fully recrystallized equiaxed FCC grains with annealing twins. (d–f) Corresponding EBSD IPF maps confirming random crystallographic texture. (g–i) Grain size histograms with descriptive statistics. Mean grain sizes range from ~10 μm to 35 μm.

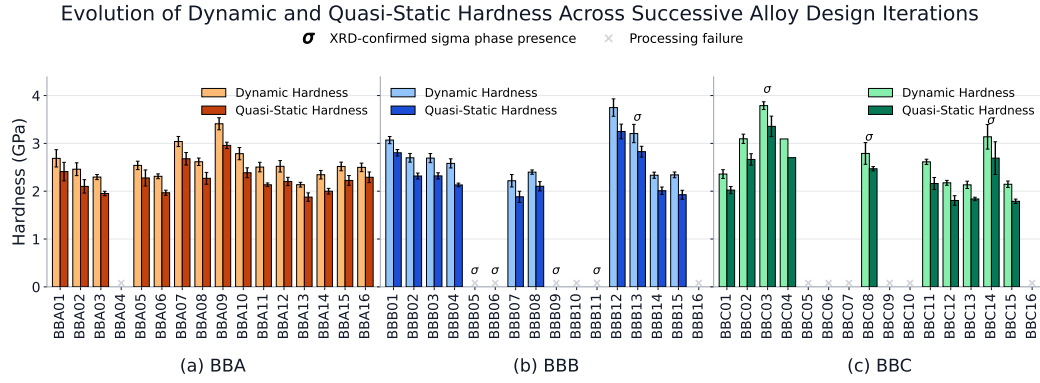


Figure 12. Dynamic and quasistatic hardness values for all candidate alloys across iterations (a) BBA, (b) BBB, and (c) BBC. Bar colors are iteration-specific within each panel. Gray $\times$ markers denote processing failures with no hardness data, and σ markers indicate alloys with XRD-confirmed sigma phase presence.

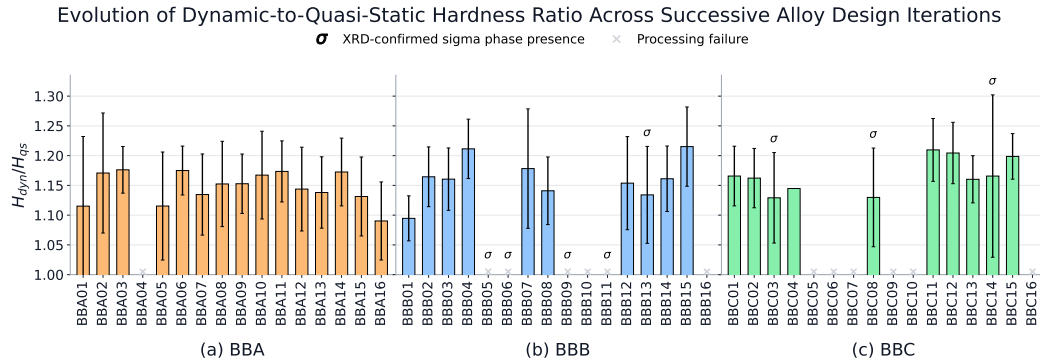


Figure 13. Dynamic-to-quasistatic hardness ratio ($H_{\text{dyn}}/H_{\text{qs}}$) for all candidate alloys across iterations (a) BBA, (b) BBB, and (c) BBC. Bar colors are iteration-specific within each panel. Gray $\times$ markers denote processing failures with no hardness-ratio data, and σ markers indicate alloys with XRD-confirmed sigma phase presence.

Evolution of Mechanical Properties and Simulated Ballistic Performance Across Iterations

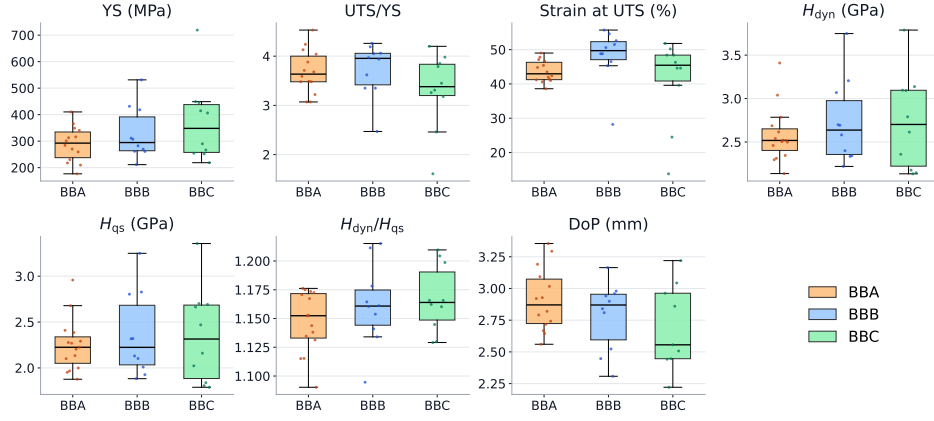


Figure 14. Evolution of mechanical properties and simulated ballistic performance across iterations. YS = yield strength, UTS = ultimate tensile strength, H_{dyn} = dynamic hardness, H_{qs} = quasistatic hardness, and DoP = simulated depth of penetration. Median strength metrics and $H_{\text{dyn}}/H_{\text{qs}}$ increase from BBA to BBC, while simulated depth of penetration decreases and strain at UTS is preserved.

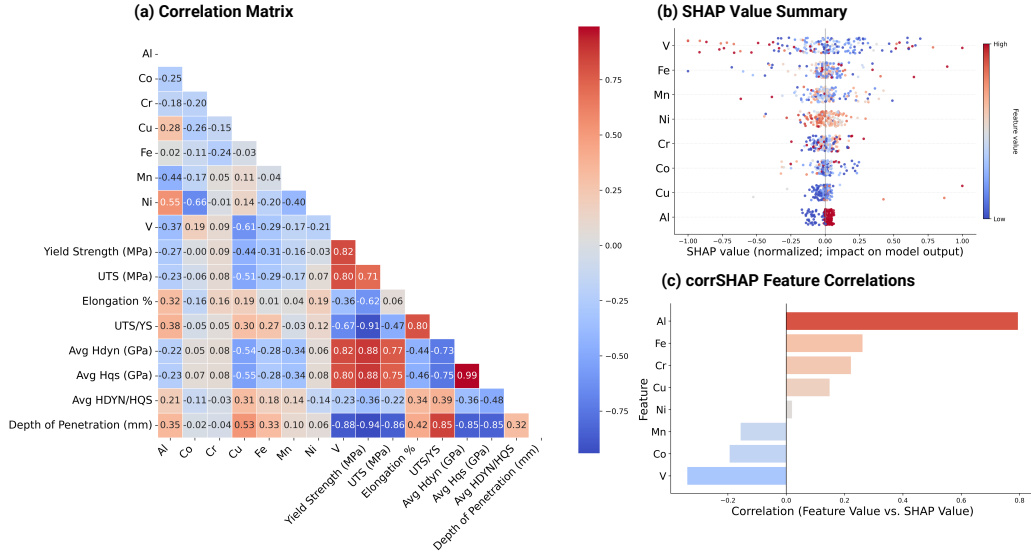


Figure 15. (a) Pearson correlation matrix among elemental concentrations, mechanical properties, and DoP. (b) SHAP beeswarm plot ranked by mean absolute SHAP value; dot color encodes feature value (red = high, blue = low). (c) corrSHAP bar chart showing the net directional influence of each element on the model output.

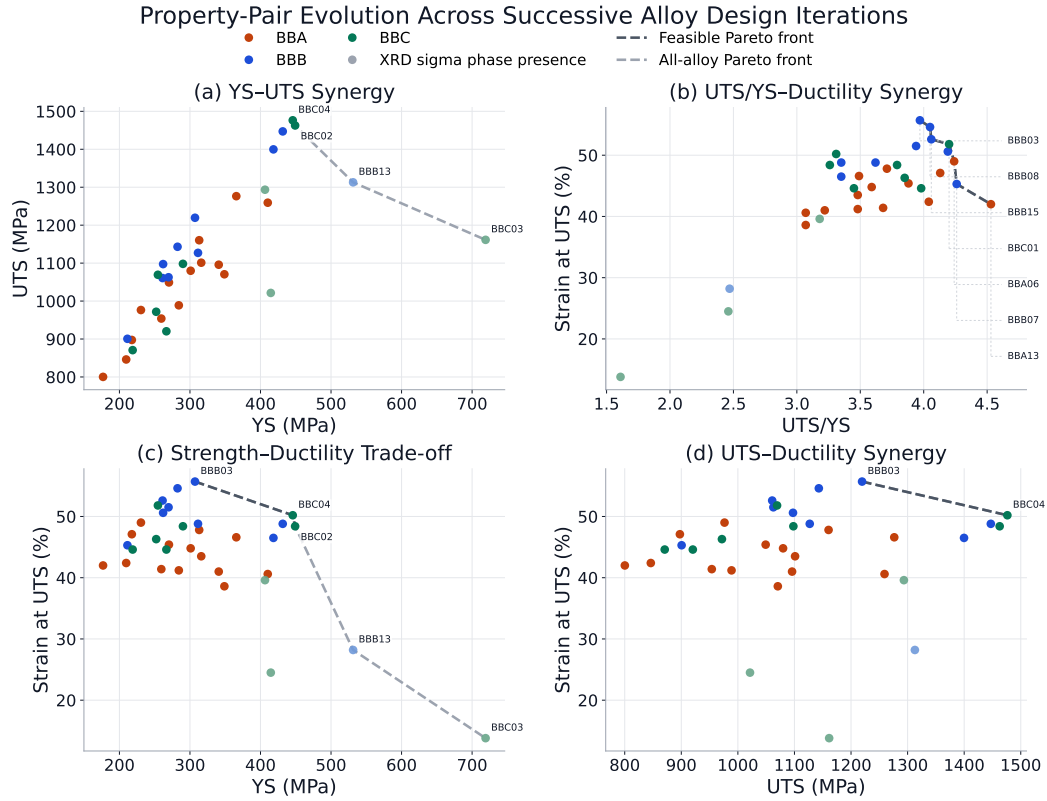


Figure 16. Pair plots of mechanical property combinations across three successive design iterations. Filled points denote single-phase FCC alloys and desaturated points indicate XRD-confirmed sigma phase presence. Dashed dark-grey lines trace the Pareto front considering only feasible alloys, while dashed light-grey lines trace the Pareto front considering all alloys.

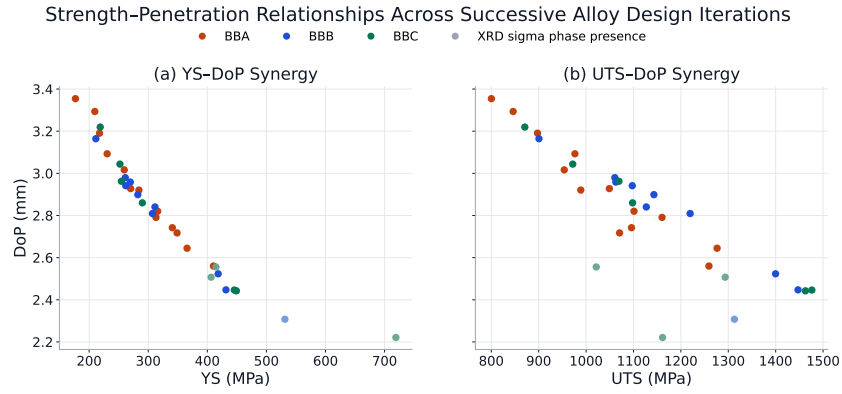


Figure 17. Simulated depth of penetration (DoP) versus quasi-static strength metrics across all tested alloys: (a) yield strength (YS) and (b) ultimate tensile strength (UTS). In both panels, higher strength is associated with lower DoP, indicating that the strongest alloys also show the greatest simulated impact resistance.

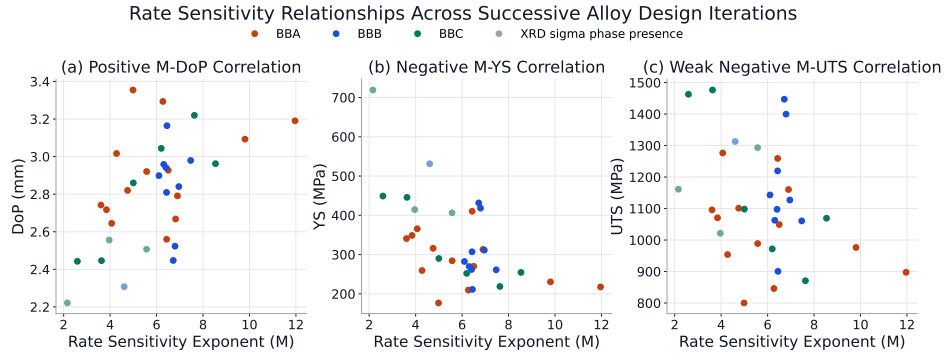


Figure 18. Rate sensitivity exponent (m) versus (a) simulated DoP, (b) yield strength, and (c) UTS for all candidate alloys across three iterations.

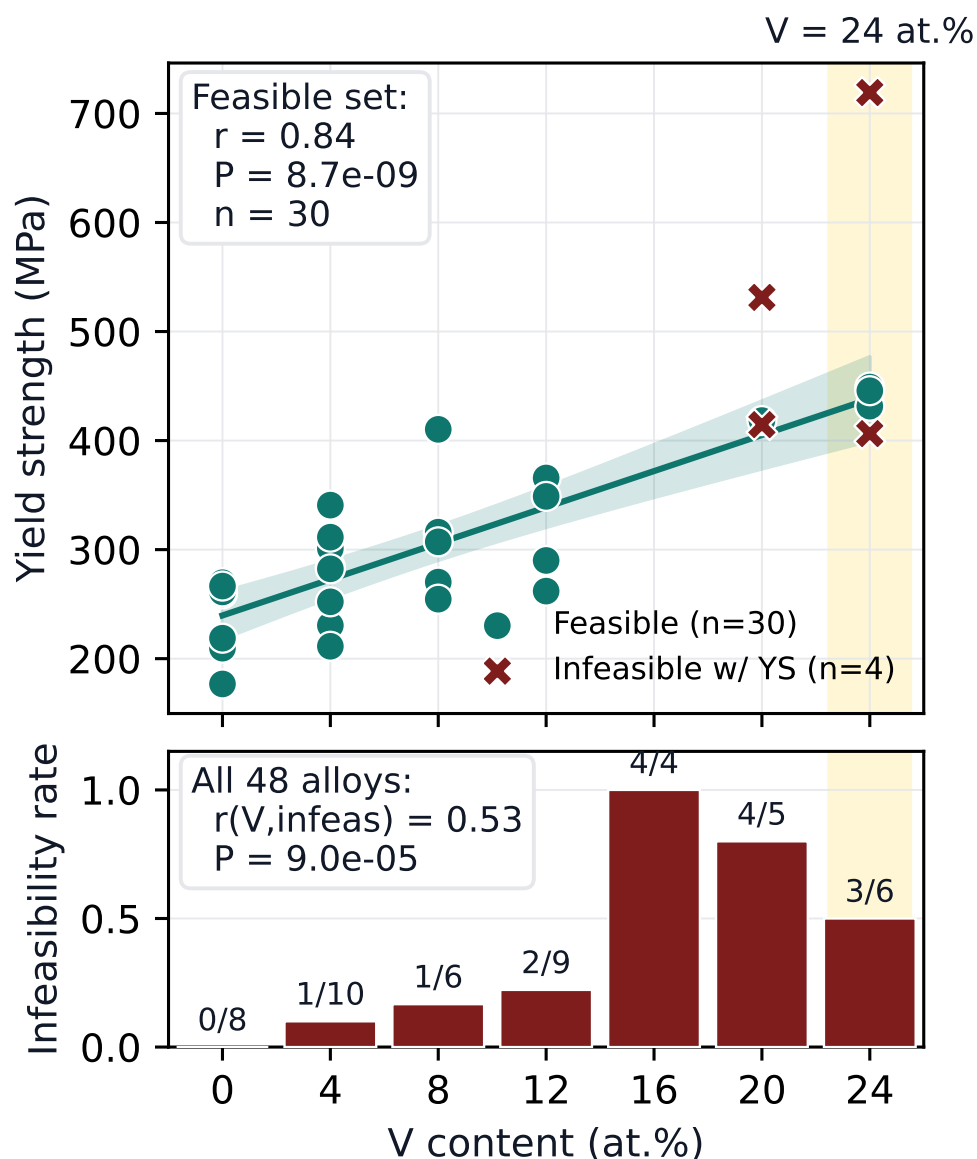


Figure 19. The two correlations that define the boundary-adjacent optimum plotted on a shared V axis. **(top)** Yield strength versus V content for the 30 single-phase FCC alloys with measurable YS (teal), with the four infeasible alloys that still yielded a YS measurement shown as maroon crosses. The regression on the feasible set gives $r = 0.84$ ($P = 8.7 \times 10^{-9}$). **(bottom)** Fraction of infeasible outcomes per V bin across all 48 alloys, with counts annotated as fails/total. The Pearson correlation between V content and infeasibility is $r = 0.54$ ($P = 9 \times 10^{-5}$). The yellow band at $V = 24$ at.% marks the compositional point at which the three strongest alloys in the campaign and three sigma failures co-occur. The non-monotonic drop in failure rate from $V = 16$ to $V = 24$ at.% reflects the BO's learned V–Ni–Cr coupling: high-V candidates sampled in later iterations were preferentially those with high Ni and low Cr, whose feasibility survives at the boundary.

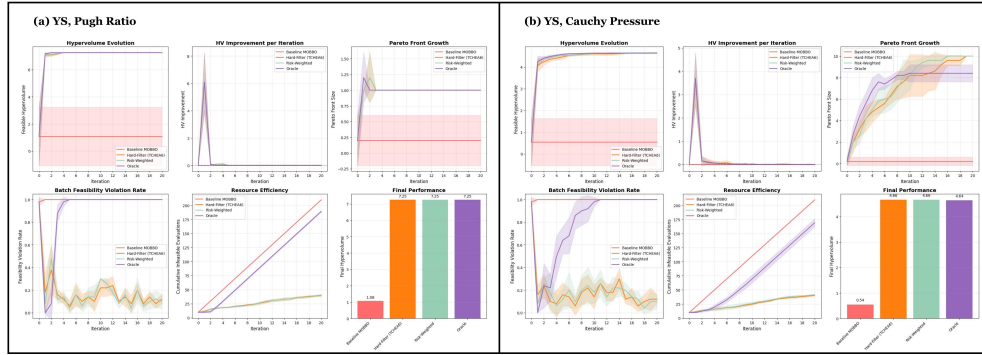


Figure 20. Benchmarking results for four MOBBO strategies on the six-component HEA system. Two bi-objective tasks are shown: (a) CV(YS)–Pugh ratio and (b) CV(YS)–Cauchy pressure. Metrics: feasible hypervolume (HV), Pareto front size, and feasibility-violation rate (FVR). Risk-weighted MOBBO matches oracle performance while maintaining low FVR.